

LINEAR ALGEBRA AND ITS APPLICATION



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Linear Algebra and Its Application

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CHAPTER OVERVIEW

0: Why Linear Algebra?

Why It Matters

Linear algebra is the study of vectors, matrices, and linear transformations—a framework for understanding relationships between quantities. It is foundational to both pure and applied mathematics, providing tools to model and solve complex problems efficiently. By mastering linear algebra, you gain the ability to think critically about structure, patterns, and systems that appear across disciplines.

Real-World Applications

Linear algebra is everywhere in science and technology:

- **Engineering & Physics:** Analyze forces, electrical circuits, and mechanical systems.
- **Computer Science:** Power graphics, machine learning algorithms, and data compression.
- **Economics & Social Sciences:** Model networks, optimize resources, and predict outcomes.
- **Everyday Life:** Recommend content on streaming platforms, analyze patterns in social networks, and understand trends in data.

What You Will Learn

By engaging with this course, you will:

1. Solve systems of linear equations using matrices and row operations.
2. Understand the structure and properties of vector spaces.
3. Apply concepts of linear independence, basis, and dimension.
4. Use eigenvalues and eigenvectors to analyze transformations and dynamic systems.
5. Connect theory to applications in physics, engineering, computer science, and beyond.

Linear algebra is not only a set of techniques—it is a way of thinking. It equips you with both the conceptual understanding and practical tools to approach complex problems with clarity and confidence. This book will guide you step by step, blending theory, examples, and applications to help you see both the beauty and utility of linear algebra.

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CHAPTER OVERVIEW

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1.1: Systems of Linear Equations

Learning Objectives

- Define linear equations and systems of linear equations.
- Classify systems of linear equations based on the existence and uniqueness of solutions.
- Interpret geometrically two-variable systems as intersecting, parallel, or coinciding lines.
- Extend geometric understanding to three-variable systems, including intersecting, non-intersecting, or coinciding planes.
- Recognize the increasing complexity of systems as the number of variables grows and the need for linear algebra techniques to analyze them.

We will start with a definition of a linear equation. You may have encountered them before, but here is a general definition that we will use.

Definition 1.1.1: Linear

A **linear equation** in the variables $x_1, x_2, \dots, x_n$ is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b \quad (1.1.1)$$

where $a_1, a_2, \dots, a_n$ are called **coefficients** and b is called **constant** all are real or complex numbers.

Example 1.1.1

Which of the following equations is a linear equation

$$3x + 4y = 2z \quad (1.1.2)$$

$$x_1x_2 + 3x_4 = x_3 \quad (1.1.3)$$

Solution

The equation $3x + 4y = 2z$ is linear since it satisfies the definition above. However $x_1x_2 + 3x_4 = x_3$ since the two variables x_1 and (x_2) are **multiplied together**.

We will usually move the unknowns to the left side of the equation, and move the constants to the right.

Definition 1.1.1: System of Linear Equations

A **system of linear equations** is a list of equations,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

where a_{ij} and b_j are real numbers. The above is a system of m equations in the n variables, $x_1, x_2, \dots, x_n$. Written more simply in terms of summation notation, the above can be written in the form

$$\sum_{j=1}^n a_{ij}x_j = b_i, \quad i = 1, 2, 3, \dots, m$$

The relative size of m and n is not important here. Notice that we have allowed a_{ij} and b_j to be any real number. We can also call these numbers **scalars**. We will use this term throughout the text, so keep in mind that the term **scalar** just means that we are working with real numbers.

✓ Example 1.1.1

Consider the system of linear equations,

$$\begin{cases} x + 2y = 3 \\ y + 2z = -3. \end{cases}$$

how many variables are involved in the system

Solution

The system consists of two equations and **three variables**, namely **x**, **y**, and **z**, since each variable appears at least once in the system.

Definition 1.1.4: $\mathbb{R}^n$

Let n be a positive whole number. We define

$$\mathbb{R}^n = \text{all ordered } n\text{-tuples of real numbers } (x_1, x_2, x_3, \dots, x_n).$$

An n -tuple of real numbers is called a **point** of $\mathbb{R}^n$.

In other words, $\mathbb{R}^n$ is just the set of all (ordered) lists of n real numbers. We will draw pictures of $\mathbb{R}^n$ in a moment, but keep in mind that *this is the definition*. For example, $(0, \frac{3}{2}, -\pi)$ and $(1, -2, 3)$ are points of $\mathbb{R}^3$. Also the point $(0, 0, 0, 0, 0)$ is in $\mathbb{R}^5$ and it has a specific name **the origin** of $\mathbb{R}^5$.

Definition 1.1.2: Solution sets

- A **solution** of a system of equations is the point $(x_1, x_2, \dots, x_n)$ that make all of the equations true simultaneously.
- The **solution set** of a system of equations is the collection of all solutions.
- **Solving** the system means finding all of the solutions.

A system of linear equations can have **one unique solution**, **infinitely many solutions**, or **no solution** at all. Below are the terminologies used to describe the two main categories of solutions for any given system of linear equations.

Definition 1.1.3: Inconsistent and Consistent

A system of equations is called **inconsistent** if it has no solutions. It is called **consistent** otherwise.

✓ Example 1.1.1

Determine whether the system is consistent or inconsistent:

$$\begin{cases} 2x - y = -1 \\ 2x - y = 3 \end{cases} \quad (1.1.4)$$

Solution:

We will use elimination to solve. First multiply the second equation by negative and then add the two equations

$$\begin{array}{rcl} 2x - y & = & -1 \\ -(2x - y & = & 3) \\ \hline 0 & = & -4 \end{array} \quad (1.1.5)$$

Conclusion: Since we arrived at a false statement, there is no point of intersection, and the system is **inconsistent**. Graphically, these lines are parallel. You can plot the two lines to confirm that they never intersect.

✓ Example 1.1.1

Determine whether the following system of equations is consistent or inconsistent. Solve the system using back-substitution.

$$\begin{cases} x + 2y + 3z = 9 \\ y + z = 3 \\ z = 1 \end{cases} \quad (1.1.6)$$

Solution

In this system, we already know the value of z , and the second equation involves only y and z . The fastest way to solve is:

1. Substitute $z = 1$ into the second equation to solve for y :

$$y + 1 = 3 \implies y = 2 \quad (1.1.7)$$

2. Substitute $y = 2$ and $z = 1$ into the first equation to solve for x :

$$x + 2(2) + 3(1) = 9 \implies x + 4 + 3 = 9 \implies x = 2 \quad (1.1.8)$$

Thus, we have the unique point in $\mathbb{R}^3$ as the solution:

$$(x, y, z) = (2, 2, 1) \quad (1.1.9)$$

Thus, the system is **consistent**.

Systems of Equations, Geometry

🧠 Outcomes

- A. Relate the types of solution sets of a system of two (three) variables to the intersections of lines in a plane (the intersection of planes in three space)

As you may remember, linear equations like $2x + 3y = 6$ can be graphed as straight lines in the coordinate plane. We say that this equation is in two variables, in this case x and y . Suppose you have two such equations, each of which can be graphed as a straight line, and consider the resulting graph of two lines. What would it mean if there exists a point of intersection between the two lines? This point, which lies on *both* graphs, gives x and y values for which both equations are true. In other words, this point gives the ordered pair (x, y) that satisfy both equations. If the point (x, y) is a point of intersection, we say that (x, y) is a **solution** to the two equations. In linear algebra, we often are concerned with finding the solution(s) to a system of equations, if such solutions exist. First, we consider graphical representations of solutions and later we will consider the algebraic methods for finding solutions.

When looking for the intersection of lines on a graph, several situations may occur. The following graphs shows the possible cases for two equations (top three graphs) and three equations (bottom two graphs), each involving two variables.

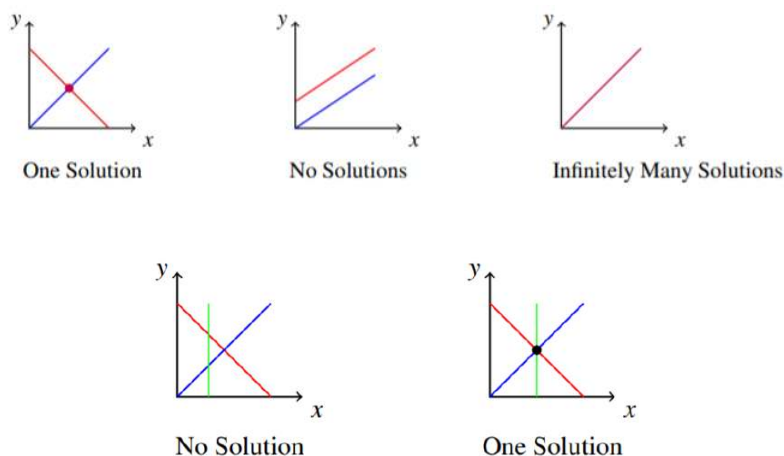


Figure 1.1.1

This illustration shows the possible solutions of a linear system. In the first case, graphs labeled “**one solution**” show lines intersecting at a single point, representing a unique solution to the system. In the second case, graphs labeled “**no solution**” show lines that do not intersect—either because they are parallel or because not all lines meet at the same point—indicating that there is no solution. In the third case, graphs labeled “**infinitely many solutions**” show lines that coincide, meaning they are the same line; here, every point on the line satisfies both equations, resulting in infinitely many solutions.

✓ Example 1.1.1: A Graphical Solution

Use a graph to find the solution to the following system of equations

$$x + y = 3$$

$$y - x = 5$$

Solution

Through graphing the above equations and identifying the point of intersection, we can find the solution(s). Remember that we must have either one solution, infinitely many, or no solutions at all. The following graph shows the two equations, as well as the intersection. Remember, the point of intersection represents the solution of the two equations, or the (x, y) which satisfy both equations. In this case, there is one point of intersection at $(-1, 4)$ which means we have one unique solution, $x = -1, y = 4$.

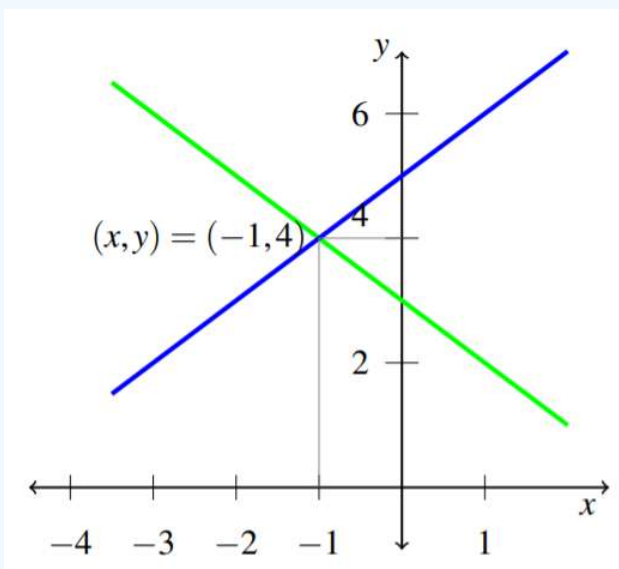


Figure 1.1.2

In the above example, we investigated the intersection point of two equations in two variables, x and y . Now we will consider the graphical solutions of three equations in two variables.

Consider the first picture above. While all three lines intersect with one another, there is no common point of intersection where all three lines meet at one point. Hence, there is no solution to the three equations. Remember, a solution is a point (x, y) which satisfies **all** three equations. In the case of the second picture, the lines intersect at a common point. This means that there is one solution to the three equations whose graphs are the given lines. You should take a moment now to draw the graph of a system which results in three parallel lines. Next, try the graph of three identical lines. Which type of solution is represented in each of these graphs?

We have now considered the graphical solutions of systems of two equations in two variables, as well as three equations in two variables. However, there is no reason to limit our investigation to equations in two variables. We will now consider equations in three variables.

You may recall that equations in three variables, such as $2x + 4y - 5z = 8$, form a plane. Above, we were looking for intersections of lines in order to identify any possible solutions. When graphically solving systems of equations in three variables, we look for intersections of planes. These points of intersection give the (x, y, z) that satisfy all the equations in the system. What types of

solutions are possible when working with three variables? Consider the following picture involving two planes, which are given by two equations in three variables.

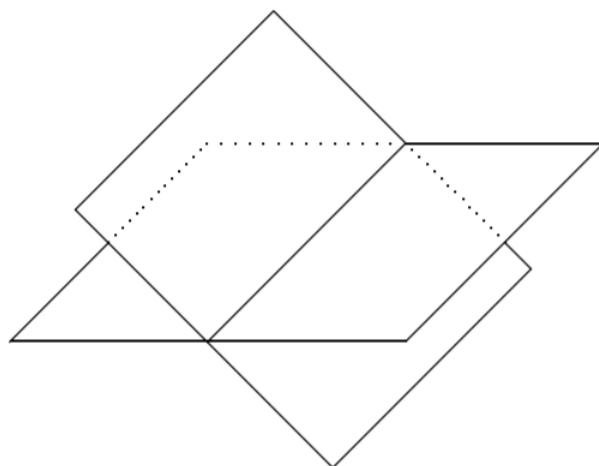


Figure 1.1.4

Notice how these two planes intersect in a line. This means that the points (x, y, z) on this line satisfy both equations in the system. Since the line contains infinitely many points, this system has infinitely many solutions.

It could also happen that the two planes fail to intersect. However, is it possible to have two planes intersect at a single point? Take a moment to attempt drawing this situation, and convince yourself that it is not possible! This means that when we have only two equations in three variables, there is no way to have a unique solution! Hence, the types of solutions possible for two equations in three variables are no solution or infinitely many solutions.

Now imagine adding a third plane. In other words, consider three equations in three variables. What types of solutions are now possible? Consider the following diagram.

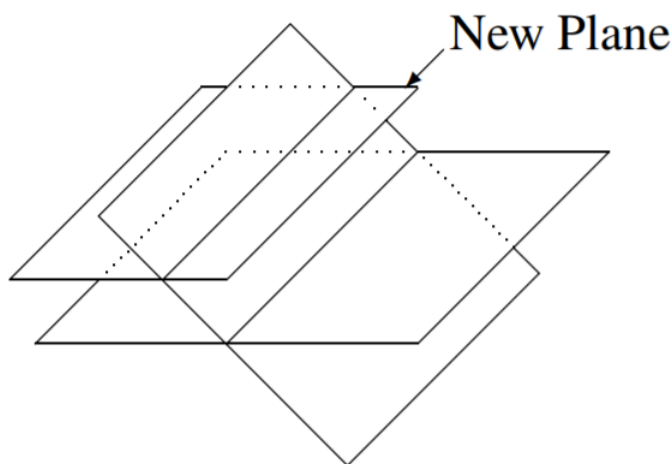


Figure 1.1.5

In this diagram, there is no point which lies in all three planes. There is no intersection between **all** planes so there is no solution. The picture illustrates the situation in which the line of intersection of the new plane with one of the original planes forms a line parallel to the line of intersection of the first two planes. However, in three dimensions, it is possible for two lines to fail to intersect even though they are not parallel. Such lines are called **skew lines**.

Recall that when working with two equations in three variables, it was not possible to have a unique solution. Is it possible when considering three equations in three variables? In fact, it is possible, and we demonstrate this situation in the following picture.

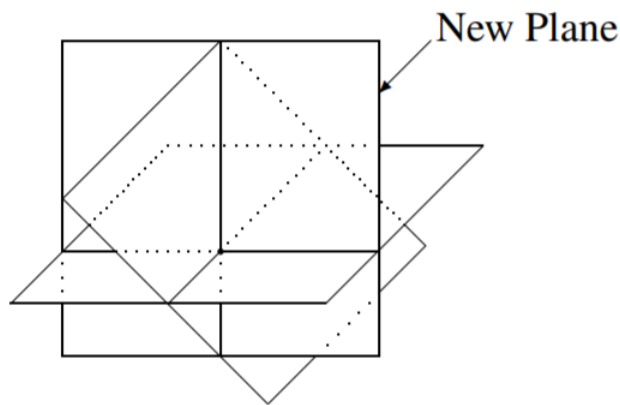


Figure 1.1.6

In this case, the three planes have a single point of intersection. Can you think of other types of solutions possible? Another is that the three planes could intersect in a line, resulting in infinitely many solutions, as in the following diagram.

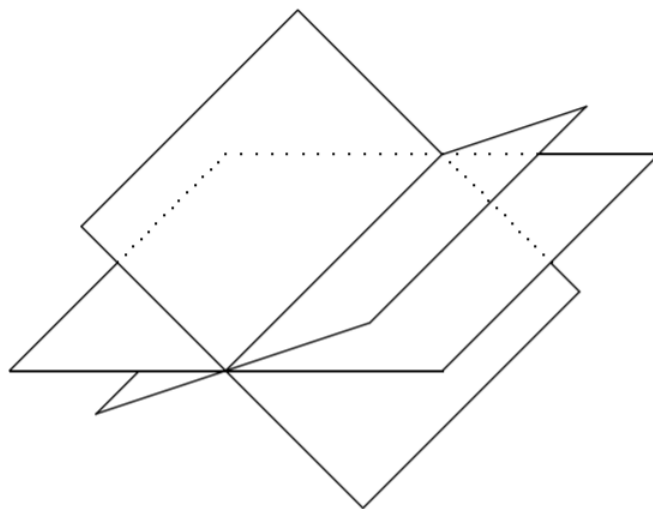


Figure 1.1.7

We have now seen how three equations in three variables can have no solution, a unique solution, or intersect in a line resulting in infinitely many solutions. It is also possible that the three equations graph the same plane, which also leads to infinitely many solutions.

You can see that when working with equations in three variables, there are many more ways to achieve the different types of solutions than when working with two variables. It may prove enlightening to spend time imagining (and drawing) many possible scenarios, and you should take some time to try a few.

You should also take some time to imagine (and draw) graphs of systems in more than three variables. Equations like $x + y - 2z + 4w = 8$ with more than three variables are often called **hyper-planes**. You may soon realize that it is tricky to draw the graphs of hyper-planes! Through the tools of linear algebra, we can algebraically examine these types of systems which are difficult to graph. In the following section, we will consider these algebraic tools.

? Exercise 1.1.1

Graphically, find the point (x_1, y_1) which lies on both lines, $x + 3y = 1$ and $4x - y = 3$. That is, graph each line and see where they intersect.

? Exercise 1.1.2

Graphically, find the point of intersection of the two lines, $3x + y = 3$ and $x + 2y = 1$. That is, graph each line and see where they intersect

? Exercise 1.1.3

You have a system of k equations in two variables, $k \geq 2$. Explain the geometric significance of

- No solution.
- A unique solution.
- An infinite number of solutions.

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1.2: Matrix Notation and Row Reduction

Learning Objectives

- Represent a system of linear equations as an augmented matrix.
- Apply elementary row operations to simplify matrices systematically.
- Transform a matrix into row echelon form (REF) and reduced row echelon form (RREF).
- Identify basic and free variables in a system and understand their implications for solutions.
- Determine whether a system has a unique solution, infinitely many solutions, or no solution.
- Express solutions with parameters when appropriate and solve practical problems using these techniques.

In the previous section, we reviewed definitions and solved systems of equations with two and three variables, much like the methods you encountered in a precalculus course. In this section, we will shift to the linear algebra perspective, where systematic tools—such as matrices and row operations—allow us to solve systems more efficiently and extend these ideas to larger and more complex systems.

Augmented Matrix

In this section, we will explore a less cumbersome way to find the solutions. First, we will represent a linear system with an **augmented matrix**. A **matrix** is simply a rectangular array of numbers. The size or dimension of a matrix is defined as where m is the number of rows and n is the number of columns. In order to construct an augmented matrix from a linear system, we create a **coefficient matrix** from the coefficients of the variables in the system, as well as a **constant matrix** from the constants. The coefficients from one equation of the system create one row of the augmented matrix.

For example, consider the linear system of three variables and three equations

$$\begin{cases} (E1) & 2x + 3y - z = 1 \\ (E2) & \quad 10x - z = 2 \\ (E3) & 4x - 9y + 2z = 5 \end{cases}$$

We encode this system into a matrix by assigning each equation to a corresponding row. Within that row, each variable and the constant gets its own column, and to separate the variables on the left hand side of the equation from the constants on the right hand side, we use a vertical bar, $|$. Note that in $E2$, since y is not present, we record its coefficient as 0. The matrix augmented this system is the following 3×3

$$\begin{array}{l} (E1) \rightarrow \\ (E2) \rightarrow \\ (E3) \rightarrow \end{array} \left(\begin{array}{ccc|c} x & y & z & b \\ 2 & 3 & -1 & 1 \\ 10 & 0 & -1 & 2 \\ 4 & -9 & 2 & 5 \end{array} \right)$$

Notice that the line separates the coefficient matrix, allowing us to study it separately. We will see that this will be useful in another representation of a linear system (see the note below).

Consider the following definition.

Definition 1.2.1: Augmented Matrix of a Linear System

For a linear system of the form

$$\begin{aligned} a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

where the x_i are variables and the a_{ij} and b_i are constants, the augmented matrix of this system is given by

$$\left(\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & & \vdots & \vdots \end{array} \right)$$

$$\left(\begin{array}{ccc|c} a_{m1} & \cdots & a_{mn} & b_m \end{array} \right)$$

We will look at more examples of how to write the corresponding augmented matrix for a system of equations, since this is the first step in solving with linear algebra methods. Afterward, we will introduce an algorithm for performing row operations on augmented matrices, which will serve as the foundation for solving systems systematically.

Elementary Row Operations

When solving systems of equations, we often perform certain operations to simplify the equations without changing their solutions. For example, we might swap two equations, multiply an equation by a nonzero constant, or replace one equation with a combination of itself and another. These same operations can be carried out directly on the rows of an augmented matrix. In this way, changes to a system of equations correspond exactly to changes in its augmented matrix, and the solution set remains unchanged. With this motivation, we now introduce the formal definition of **elementary row operations**, which will be our main tool for solving systems systematically.

Definition 1.2.2: Elementary Row Operations

The **elementary row operations** (also known as **row operations**) consist of the following

1. Switch two rows.
2. Multiply a row by a nonzero number.
3. Replace a row by any multiple of another row added to it.

We first introduce an example to demonstrate **row operations** applied to a system of linear equations and their corresponding **augmented matrix**. Below are some possible row operations that can be used to solve the system. Notice that the **goal is to eliminate variables systematically** to find the solution, and the primary focus here is to **understand the notation** for these operations.

✓ Example 1.2.1

Consider the following system of linear equations:

$$\begin{cases} 3x - y + z = 3 \\ 2x - 4y + 3z = 16 \\ x - y + z = 5 \end{cases} \quad (1.2.1)$$

We start by forming an augmented matrix:

$$\begin{cases} 3x - y + z = 3 \\ 2x - 4y + 3z = 16 \\ x - y + z = 5 \end{cases} \xrightarrow{\text{augmented matrix}} \left(\begin{array}{ccc|c} 3 & -1 & 1 & 3 \\ 2 & -4 & 3 & 16 \\ 1 & -1 & 1 & 5 \end{array} \right) \quad (1.2.2)$$

Below are possible row operations applied to the system and their corresponding augmented matrices. Notice that we use a single arrow *rightarrow* to apply a row operation to one row, and a double arrow *leftrightarrow* when swapping rows.

-Switch two rows.:

$$\begin{cases} (E1) & 3x - y + z = 3 \\ (E2) & 2x - 4y + 3z = 16 \\ (E3) & x - y + z = 5 \end{cases} \xrightarrow{\text{Switch } E1 \text{ and } E3} \begin{cases} (E1) & x - y + z = 5 \\ (E2) & 2x - 4y + 3z = 16 \\ (E3) & 3x - y + z = 3 \end{cases}$$

$$\left(\begin{array}{ccc|c} 3 & -1 & 1 & 3 \\ 2 & -4 & 3 & 16 \\ 1 & -1 & 1 & 5 \end{array} \right) \xrightarrow{R_1 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 5 \\ 2 & -4 & 3 & 16 \\ 3 & -1 & 1 & 3 \end{array} \right)$$

-Multiply a row by a nonzero number.

$$\begin{cases} (E1) & 3x_1 + x_2 + x_4 = 6 \\ (E2) & 2x_1 - 4x_2 + 3x_3 + x_4 = 16 \\ (E3) & x_1 - x_2 + x_3 + x_4 = 5 \end{cases} \xrightarrow{R_1 \leftarrow \frac{1}{3}R_1} \begin{cases} (E1) & x_1 + \frac{1}{3}x_2 + \frac{1}{3}x_4 = 2 \\ (E2) & 2x_1 - 4x_2 + 3x_3 + x_4 = 16 \\ (E3) & x_1 - x_2 + x_3 + x_4 = 5 \end{cases}$$

$$\left\{ \begin{array}{l} (E2) \quad 2x_1 + x_2 - x_3 = 4 \\ (E3) \quad x_2 - 3x_3 - 2x_4 = 0 \end{array} \right. \xrightarrow{\text{Replace } E1 \text{ with } \frac{1}{3}E1} \left\{ \begin{array}{l} (E2) \quad 2x_1 + x_2 - x_3 = 4 \\ (E3) \quad x_2 - 3x_3 - 2x_4 = 0 \end{array} \right.$$

$$\left(\begin{array}{cccc|c} 3 & 1 & 0 & 1 & 6 \\ 2 & 1 & -1 & 0 & 4 \\ 0 & 1 & -3 & -2 & 0 \end{array} \right) \xrightarrow{R_1 \rightarrow \frac{1}{3}R_1} \left(\begin{array}{cccc|c} 1 & \frac{1}{3} & 0 & \frac{1}{3} & 2 \\ 2 & 1 & -1 & 0 & 4 \\ 0 & 1 & -3 & -2 & 0 \end{array} \right)$$

-Replace a row by any multiple of another row added to it.

$$\left\{ \begin{array}{l} (E1) \quad x + \frac{3}{2}y - \frac{1}{2}z = \frac{1}{2} \\ (E2) \quad 10x - z = 2 \\ (E3) \quad 4x - 9y + 2z = 5 \end{array} \right. \xrightarrow{\begin{array}{l} \text{Replace } E2 \text{ with } -10E1+E2 \\ \text{Replace } E3 \text{ with } -4E1+E3 \end{array}} \left\{ \begin{array}{l} (E1) \quad x + \frac{3}{2}y - \frac{1}{2}z = \frac{1}{2} \\ (E2) \quad -15y + 4z = -3 \\ (E3) \quad -15y + 4z = 3 \end{array} \right.$$

$$\left(\begin{array}{ccc|c} 1 & \frac{3}{2} & -\frac{1}{2} & \frac{1}{2} \\ 10 & 0 & -1 & 2 \\ 4 & -9 & 2 & 5 \end{array} \right) \xrightarrow{\begin{array}{l} R_2 \rightarrow -10R_1 + R_2 \\ R_3 \rightarrow -4R_1 + R_3 \end{array}} \left(\begin{array}{ccc|c} 1 & \frac{3}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & -15 & 4 & -3 \\ 0 & -15 & 4 & 3 \end{array} \right)$$

To solve a system of linear equations, we can **translate elementary row operations into an elimination method** applied to the augmented matrix of the system. Elementary row operations—swapping rows, multiplying a row by a nonzero scalar, and adding a multiple of one row to another—allow us to systematically **eliminate variables** from equations. By performing these operations strategically, we can transform the augmented matrix into an **upper triangular (staircase) form**, where each successive row contains fewer variables. This triangular form makes the system easier to solve using **back substitution**, starting from the last equation and substituting values upward to find all unknowns. This method not only streamlines computations but also clearly demonstrates the structure and dependencies within the system.

Below is an example showing how to apply row operations to solve a system of linear equations using the elimination method.

✓ Example 1.2.2

Solve the following system of linear equation:

$$\left\{ \begin{array}{l} x + 2y + 3z = 6 \\ 2x - 3y + 2z = 14 \\ 3x + y - z = -2 \end{array} \right. \quad (1.2.3)$$

Solution

We start by forming an augmented matrix:

$$\left\{ \begin{array}{l} x + 2y + 3z = 6 \\ 2x - 3y + 2z = 14 \\ 3x + y - z = -2 \end{array} \right. \xrightarrow{\text{augmented matrix}} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 2 & -3 & 2 & 14 \\ 3 & 1 & -1 & -2 \end{array} \right).$$

Eliminating a variable from an equation means producing a zero to the left of the line in an augmented matrix. First we produce zeros in the first column (i.e. we eliminate x) by subtracting multiples of the first row.

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 2 & -3 & 2 & 14 \\ 3 & 1 & -1 & -2 \end{array} \right) \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & -7 & -4 & 2 \\ 3 & 1 & -1 & -2 \end{array} \right)$$

$$\xrightarrow{R_3 \rightarrow R_3 - 3R_1} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & -7 & -4 & 2 \\ 0 & -5 & -10 & -20 \end{array} \right) \quad (1.2.4)$$

This was made much easier by the fact that the top-left entry is equal to 1, so we can simply multiply the first row by the number below and subtract. In order to eliminate y in the same way, we would like to produce a 1 in the second column. We could divide the second row by -7 , but this would produce fractions; instead, let's divide the third by -5 .

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & -7 & -4 & 2 \\ 0 & -5 & -10 & -20 \end{array} \right) \xrightarrow{R_3 \rightarrow (1/5)R_3} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & -7 & -4 & 2 \\ 0 & -1 & -2 & -4 \end{array} \right)$$

$$\begin{aligned}
 & \left(\begin{array}{ccc|c} 0 & -5 & -10 & -20 \end{array} \right) \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 0 & 1 & 2 & 4 \\ 1 & 2 & 3 & 6 \\ 0 & -7 & -4 & 2 \end{array} \right) \\
 & \xrightarrow{R_3 \rightarrow R_3 + 7R_2} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 10 & 30 \end{array} \right) \\
 & \xrightarrow{R_3 \rightarrow R_3 \div 10} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & 3 \end{array} \right)
 \end{aligned} \tag{1.2.5}$$

We swapped the second and third row just to keep things orderly. Now we translate this augmented matrix back into a system of equations:

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 1 & 3 \end{array} \right) \xrightarrow{\text{becomes}} \begin{cases} x + 2y + 3z = 6 \\ y + 2z = 4 \\ z = 3 \end{cases}$$

Hence $z = 3$; back-substituting as in Example 1.2.1 gives $(x, y, z) = (1, -2, 3)$.

Performing row operations on a matrix **does not change the solution set** of the corresponding system of linear equations. In other words, no matter how we swap rows, scale a row by a nonzero constant, or add a multiple of one row to another, the set of solutions remains exactly the same.

Definition 1.2.1: Row Equivalent

Two matrices are called **row equivalent** if one can be obtained from the other by doing some number of row operations.

So the linear equations of row-equivalent matrices have the *same solution set*.

✓ Example 1.2.3: An Inconsistent System

Solve the following system of equations using row operations:

$$\begin{cases} x + y = 2 \\ 3x + 4y = 5 \\ 4x + 5y = 9 \end{cases}$$

Solution

$$\begin{aligned}
 & \left(\begin{array}{cc|c} 2 & 10 & -1 \\ 3 & 15 & 2 \end{array} \right) \rightarrow R_1 \leftarrow R_1 \div 2 \rightarrow \left(\begin{array}{cc|c} 1 & 5 & -\frac{1}{2} \\ 3 & 15 & 2 \end{array} \right) \\
 & \rightarrow R_2 \leftarrow R_2 - 3R_1 \rightarrow \left(\begin{array}{cc|c} 1 & 5 & -\frac{1}{2} \\ 0 & 0 & \frac{7}{2} \end{array} \right) \\
 & \rightarrow R_2 \leftarrow R_2 \times \frac{2}{7} \rightarrow \left(\begin{array}{cc|c} 1 & 5 & -\frac{1}{2} \\ 0 & 0 & 1 \end{array} \right) \\
 & \rightarrow R_1 \leftarrow R_1 + \frac{1}{2}R_2 \rightarrow \left(\begin{array}{cc|c} 1 & 5 & 0 \\ 0 & 0 & 1 \end{array} \right)
 \end{aligned} \tag{1.2.6}$$

This row-reduced matrix corresponds to the inconsistent system:

$$\begin{cases} x + 5y = 0 \\ 0 = 1 \end{cases} \tag{1.2.7}$$

Notice the last column is a pivot column.

This translates back into the system of equations

$$\begin{cases} x + y = 2 \\ y = -1 \\ 0 = 2. \end{cases}$$

Our original system has the same solution set as this system. But this system has no solutions: there are no values of x, y making the third equation true! We conclude that our original equation was inconsistent.

Echelon Forms

In the previous examples, we saw how to translate a system of linear equations into an augmented matrix and apply row operations. The next step is to work directly with matrices, without referencing the original equations. To do this, we need a systematic algorithm for solving an augmented matrix. First, we define what it means for a matrix to be “solved.” We say a matrix is in **row echelon form (REF)** when it has a staircase-like structure: each leading entry (the first nonzero number from the left) of a row is to the right of the leading entry in the row above, and all entries below a leading entry are zero. Transforming a matrix into row echelon form using **elementary row operations** provides a clear pathway for solving the system through **back substitution**, allowing us to determine all unknowns systematically and efficiently.

Below is the official definition of row echelon form (REF):

Definition 1.2.2: Row Echelon Form

A matrix is in **row echelon form** if:

1. All zero rows are at the bottom.
2. The first nonzero entry of a row is to the *right* of the first nonzero entry of the row above.
3. Below the first nonzero entry of a row, all entries are zero.

Here is a picture of a matrix in row echelon form:

$$\begin{pmatrix} \boxed{\star} & \star & \star & \star & \star \\ 0 & \boxed{\star} & \star & \star & \star \\ 0 & 0 & 0 & \boxed{\star} & \star \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{array}{l} \star = \text{any number} \\ \boxed{\star} = \text{any nonzero number} \end{array}$$

A matrix in row-echelon form is generally easy to solve using back-substitution. For example,

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 10 & 30 \end{array} \right) \xrightarrow{\text{becomes}} \begin{cases} x + 2y + 3z = 6 \\ y + 2z = 4 \\ 10z = 30. \end{cases}$$

We immediately see that $z = 3$, which implies $y = 4 - 2 \cdot 3 = -2$ and $x = 6 - 2(-2) - 3 \cdot 3 = 1$. See Example 1.2.3.

Once a matrix is in **row echelon form (REF)**, we can take the process one step further and transform it into **reduced row echelon form (RREF)**. In RREF, not only does the matrix maintain the staircase-like structure of REF, but each leading entry is also **1** and is the only nonzero entry in its column. By continuing elimination in this way, we can solve the system **directly from the matrix**, without the need for back substitution. The goal is to simplify the matrix so that each row corresponds to a single variable with its value, providing a clear and immediate solution to the system. This approach extends the elimination method fully to a systematic, straightforward procedure for solving any system of linear equations.

Definition 1.2.4: Reduced Row Echelon Form

A matrix is in **reduced row echelon form** if it is in row echelon form, and in addition:

4. Each pivot is equal to 1.
5. Each pivot is the only nonzero entry in its column.

Here is a picture of a matrix in reduced row echelon form:

$$\begin{pmatrix} \textcolor{red}{1} & 0 & \star & 0 & \star \\ 0 & \textcolor{red}{1} & \star & 0 & \star \\ 0 & 0 & 0 & \textcolor{red}{1} & \star \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{array}{l} \star = \text{any number} \\ \textcolor{red}{1} = \text{leading term} \end{array}$$

A matrix in reduced row echelon form is in some sense completely solved. For example,

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{array} \right) \xrightarrow{\text{becomes}} \begin{cases} x = 1 \\ y = -2 \\ z = 3. \end{cases}$$

✓ Example 1.2.4

The following matrices are in **reduced row echelon form**:

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{pmatrix} \quad \begin{pmatrix} 0 & 1 & 8 & 0 \end{pmatrix} \quad \left(\begin{array}{cc|c} 1 & 17 & 0 \\ 0 & 0 & 1 \end{array} \right) \quad \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The following matrices are in row echelon form but **not reduced row echelon form**:

$$\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \quad \left(\begin{array}{ccc|c} 2 & 7 & 1 & 4 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 3 \end{array} \right) \quad \begin{pmatrix} 1 & 17 & 0 \\ 0 & 1 & 1 \end{pmatrix} \quad \begin{pmatrix} 2 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}.$$

The following matrices are **not in echelon form**:

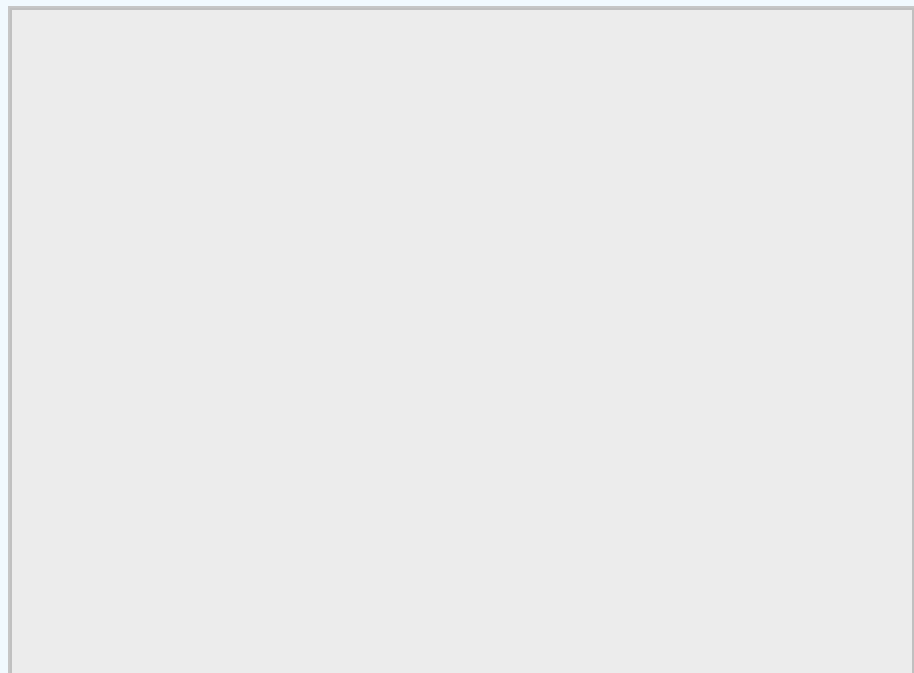
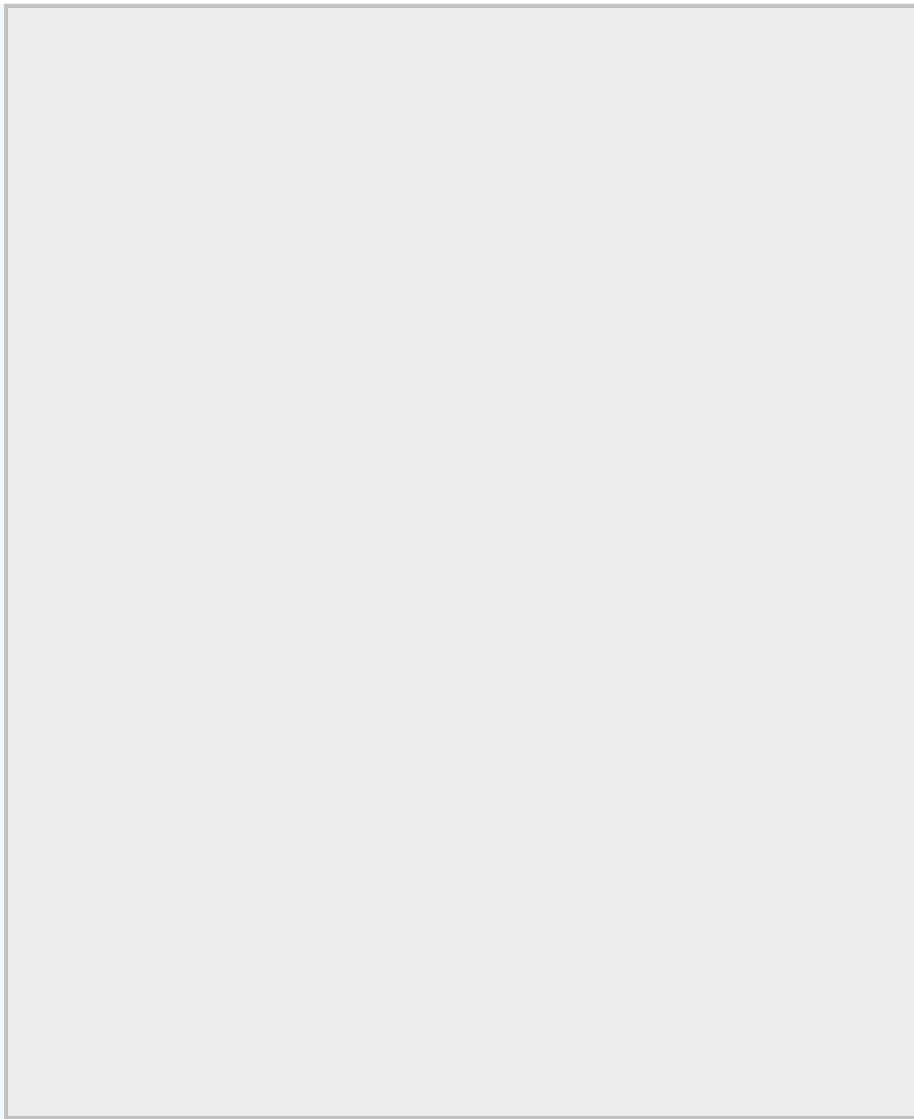
$$\left(\begin{array}{ccc|c} 2 & 7 & 1 & 4 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 3 \end{array} \right) \quad \begin{pmatrix} 0 & 17 & 0 \\ 0 & 2 & 1 \end{pmatrix} \quad \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

✓ Example 1.2.5

Reduce the following augmented matrix to solve for the corresponding system of linear equation:

$$\left(\begin{array}{ccc|c} 0 & -7 & -4 & 2 \\ 2 & 4 & 6 & 12 \\ 3 & 1 & -1 & -2 \end{array} \right).$$

Solution



The reduced row echelon form of the matrix is

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{array} \right) \xrightarrow{\text{translates to}} \begin{cases} x & = & 1 \\ y & = & -2 \\ z & = & 3. \end{cases}$$

The reduced row echelon form of the matrix tells us that the only solution is $(x, y, z) = (1, -2, 3)$.

Figure 1.2.2: Animated slideshow of the row reduction in this example.

Here is the row reduction algorithm, summarized in pictures.

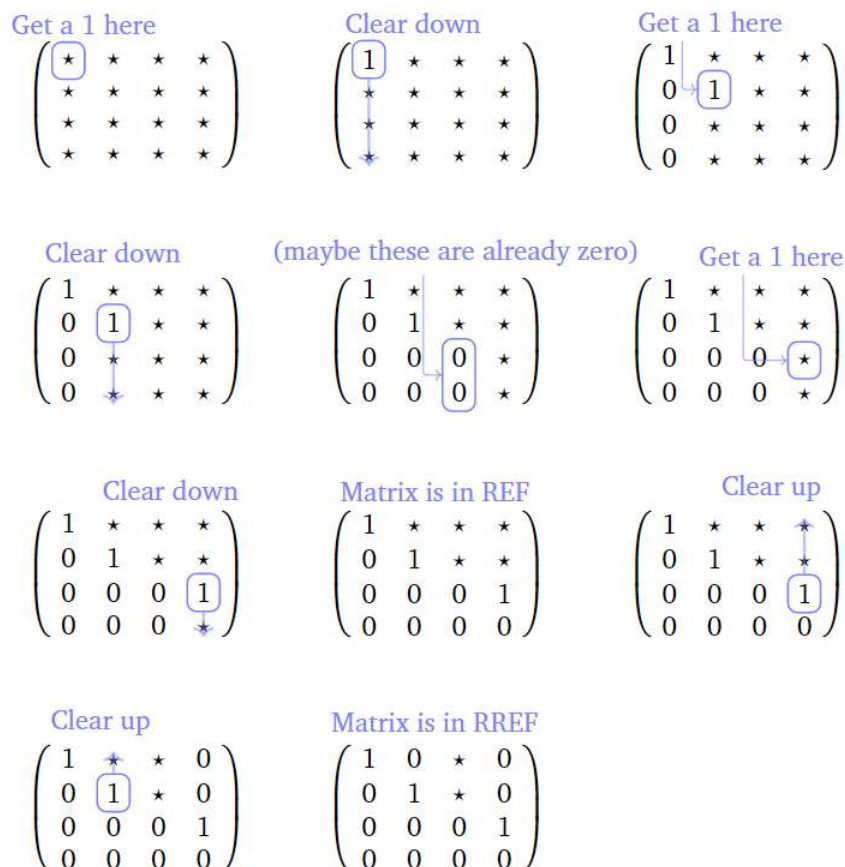


Figure 1.2.3

After working through examples of transforming a matrix into reduced row echelon form, we may begin to wonder whether the final result depends on the sequence of row operations we choose. Different paths can certainly be taken, but the remarkable fact is that all lead to the same destination. This observation is formalized in the following theorem, which guarantees the **uniqueness of the reduced echelon form** of a matrix.

Theorem 1.2.2: Uniqueness of the Reduced Row-Echelon Form

Every matrix A is equivalent to a unique matrix in reduced row-echelon form.

According to this theorem we can say that each matrix A has a unique reduced row-echelon form.

Pivot Position, Basic and Free variables

Once we obtain a matrix in reduced row echelon form (RREF), it becomes much easier to analyze the structure of the solutions. Recognizing the columns that contain leading entries is important because they correspond to the variables determined directly by the system. We call these positions **pivot positions**, and they provide a systematic way to describe the solution set of the system.

Now, looking back at the example above, notice that our goal is to create **pivot columns** and identify the corresponding **pivot positions**.

Definition 1.2.5

A **pivot position** of a matrix is an entry that is a pivot of a row echelon form of that matrix.

A **pivot column** of a matrix is a column that contains a pivot position.

✓ Example 1.2.6: Pivot Positions

Find the pivot positions and pivot columns of this matrix

$$A = \left(\begin{array}{ccc|c} 0 & -7 & -4 & 2 \\ 2 & 4 & 6 & 12 \\ 3 & 1 & -1 & -2 \end{array} \right).$$

Solution

We saw in Example 1.2.5 that a row echelon form of the matrix is

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 10 & 30 \end{array} \right).$$

The pivot positions of A are the entries that become pivots in a row echelon form; they are marked in red below:

$$\left(\begin{array}{ccc|c} 0 & -7 & -4 & 2 \\ 2 & 4 & 6 & 12 \\ 3 & 1 & -1 & -2 \end{array} \right).$$

The first, second, and third columns are pivot columns.

Once we have identified the pivot positions in the reduced row echelon form of a matrix, we can use them to describe the solutions of a system of linear equations. Pivot positions correspond to the variables that are directly determined by the system, while the remaining variables—those without pivots—are called **free variables**. The relationship between pivot and free variables provides a systematic way to express the solution set, whether it consists of a single unique solution or infinitely many. However, not all systems have solutions.

In other words, the row reduced matrix of an inconsistent system looks like this:

$$\left(\begin{array}{cccc|c} 1 & 0 & * & * & 0 \\ 0 & 1 & * & * & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right)$$

In the next example, we will see how an inconsistent system can be recognized from its reduced form and the arrangement of pivot columns.

✓ Example 1.2.7: An Inconsistent System

Solve the linear system

$$\begin{cases} 2x + 10y = -1 \\ 3x + 15y = 2 \end{cases}$$

using row reduction.

Solution

$$\left(\begin{array}{cc|c} 2 & 10 & -1 \end{array} \right) \xrightarrow{R_1 \leftarrow R_1 \div 2} \left(\begin{array}{cc|c} 1 & 5 & -\frac{1}{2} \end{array} \right)$$

$$\begin{aligned}
 & \left(\begin{array}{cc|c} 3 & 15 & 2 \end{array} \right) \xrightarrow{R_2 \leftarrow R_2 - 3R_1} \left(\begin{array}{cc|c} 1 & 5 & -\frac{1}{2} \\ 0 & 0 & \frac{7}{2} \end{array} \right) \\
 & \xrightarrow{R_2 \leftarrow R_2 \times \frac{2}{7}} \left(\begin{array}{cc|c} 1 & 5 & -\frac{1}{2} \\ 0 & 0 & 1 \end{array} \right) \\
 & \xrightarrow{R_1 \leftarrow R_1 + \frac{1}{2}R_2} \left(\begin{array}{cc|c} 1 & 5 & 0 \\ 0 & 0 & 1 \end{array} \right)
 \end{aligned} \tag{1.2.8}$$

This row-reduced matrix corresponds to the inconsistent system:

$$\begin{cases} x + 5y = 0 \\ 0 = 1 \end{cases} \tag{1.2.9}$$

In the above example, we saw how recognizing the reduced row echelon form of an inconsistent system leads us to the following theorem:

Theorem 1.2.2: The Row Echelon Form of an Inconsistent System

An augmented matrix corresponds to an inconsistent system of equations if and only if the last column (i.e., the augmented column) is a *pivot column*.

Not having a pivot column in every column—or equivalently, not having enough independent equations to assign a pivot to each variable—means that some variables will not be determined directly by the system. These undetermined variables are called free variables, as mentioned previously. The variables corresponding to pivot columns, which are determined directly by the system, are called **basic variables**. The presence of free variables indicates that the system does not have a unique solution; instead, it has infinitely many solutions. By expressing the basic (pivot) variables in terms of these free variables, we can systematically describe all possible solutions and understand the structure of the solution set.

Here are some example

✓ Example 1.2.8: A System with Many Solutions

Solve the linear system

$$\begin{cases} 2x + y + 12z = 1 \\ x + 2y + 9z = -1 \end{cases}$$

using row reduction, while identifying pivot positions, basic variables, and free variables.

Solution:

$$\begin{aligned}
 & \left(\begin{array}{ccc|c} 2 & 1 & 12 & 1 \\ 1 & 2 & 9 & -1 \end{array} \right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|c} 1 & 2 & 9 & -1 \\ 2 & 1 & 12 & 1 \end{array} \right) \quad (\text{Step 1: first pivot}) \\
 & \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 2 & 9 & -1 \\ 0 & -3 & -6 & 3 \end{array} \right) \quad (\text{Step 2: eliminate below pivot}) \\
 & \xrightarrow{R_2 \rightarrow (-1/3)R_2} \left(\begin{array}{ccc|c} 1 & 2 & 9 & -1 \\ 0 & 1 & 2 & -1 \end{array} \right) \quad (\text{Step 3: scale second pivot}) \\
 & \xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left(\begin{array}{ccc|c} 1 & 0 & 5 & 1 \\ 0 & 1 & 2 & -1 \end{array} \right) \quad (\text{Step 4: eliminate above pivot})
 \end{aligned} \tag{1.2.10}$$

This row-reduced matrix corresponds to the linear system

$$\begin{cases} x + 5z = 1 \\ y + 2z = -1 \end{cases}$$

Analysis of solutions:

- **Pivot positions:** The first entry in row 1 (column 1) and the first entry in row 2 (column 2) are pivot positions.

- **Basic variables:** The variables corresponding to pivot columns, x and y , are basic variables.
- **Free variable:** Column 3 does not contain a pivot, so z is a free variable.
- **Infinitely many solutions:** Because there is a free variable, the system has infinitely many solutions. Expressing the basic variables in terms of the free variable z :
 $x = 1 - 5z, y = -1 - 2z, z = z \text{ (free)}.$

To describe all solutions of a system with free variables more conveniently, we can introduce a parameter (a parameter represents any real number) , often denoted by t , to represent the free variable. This allows us to write the basic variables explicitly in terms of t , giving a single, unified expression for every solution in the solution set. By parametrizing the free variable, we can clearly see how the free variable affects the values of the basic variables and conveniently describe the infinitely many solutions in a compact form.

Continuing with the previous example, we can let the free variable z be represented by the parameter t . Then the solution set can be written as:

$$x = 1 - 5t, \quad y = -1 - 2t, \quad z = t, \quad t \in \mathbb{R}. \quad (1.2.11)$$

This parametrization captures all infinitely many solutions of the system, showing how each choice of t gives a corresponding set of values for the basic variables x and y .

✓ Example 1.2.1: Basic and Free Variables

Find the basic and free variables in the system

$$\begin{aligned} x + 2y - z + w &= 3 \\ x + y - z + w &= 1 \\ x + 3y - z + w &= 5 \end{aligned}$$

Solution

The row-echelon form of the augmented matrix of this system is given by

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 3 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

You can see that columns 1 and 2 are pivot columns. These columns correspond to variables x and y , making these the basic variables. Columns 3 and 4 are not pivot columns, which means that z and w are free variables.

We can write the solution to this system as

$$\begin{aligned} x &= -1 + s - t \\ y &= 2 \\ z &= s \\ w &= t \end{aligned}$$

Here the free variables are written as parameters, and the basic variables are given by linear functions of these parameters.

In general,

✓ Example 1.2.7: An Infinite Set of Solutions

Give the complete solution to the system of equations

$$\begin{aligned} 3x - y - 5z &= 9 \\ y - 10z &= 0 \\ -2x + y &= -6 \end{aligned} \quad (1.2.12)$$

Solution

The augmented matrix of this system is

$$\left[\begin{array}{ccc|c} 3 & -1 & -5 & 9 \\ 0 & 1 & -10 & 0 \\ -2 & 1 & 0 & -6 \end{array} \right]$$

In order to find the solution to this system, we will carry the augmented matrix to reduced row-echelon form. The first column is the first pivot column. We want to use row operations to create zeros beneath the first entry in this column, which is in the first pivot position. Replace the third row with 2 times the first row added to 3 times the third row. This gives

$$\left[\begin{array}{ccc|c} 3 & -1 & -5 & 9 \\ 0 & 1 & -10 & 0 \\ 0 & 1 & -10 & 0 \end{array} \right]$$

Now, we have created zeros beneath the 3 in the first column, so we move on to the second pivot column (which is the second column) and repeat the procedure. Take -1 times the second row and add to the third row.

$$\left[\begin{array}{ccc|c} 3 & -1 & -5 & 9 \\ 0 & 1 & -10 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

The entry below the pivot position in the second column is now a zero. Notice that we have no more pivot columns because we have only two leading entries.

At this stage, we also want the leading entries to be equal to one. To do so, divide the first row by 3.

$$\left[\begin{array}{ccc|c} 1 & -\frac{1}{3} & -\frac{5}{3} & 3 \\ 0 & 1 & -10 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

This matrix is now in row-echelon form.

Let's continue with row operations until the matrix is in reduced row-echelon form. This involves creating zeros above the pivot positions in each pivot column. This requires only one step, which is to add $\frac{1}{3}$ times the second row to the first row.

$$\left[\begin{array}{ccc|c} 1 & 0 & -5 & 3 \\ 0 & 1 & -10 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

This is in reduced row-echelon form, which you should verify using [Definition 1.2.4](#). The equations corresponding to this reduced row-echelon form are

$$\begin{aligned} x - 5z &= 3 \\ y - 10z &= 0 \end{aligned}$$

or

$$\begin{aligned} x &= 3 + 5z \\ y &= 10z \end{aligned}$$

Observe that z is not restrained by any equation. In fact, z can equal any number. For example, we can let $z = t$, where we can choose t to be any number. In this context t is called a **parameter**. Therefore, the solution set of this system is

$$\begin{aligned} x &= 3 + 5t \\ y &= 10t \\ z &= t \end{aligned}$$

where t is arbitrary. The system has an infinite set of solutions which are given by these equations. For any value of t we select, x , y , and z will be given by the above equations. For example, if we choose $t = 4$ then the corresponding solution would be

$$\begin{aligned} x &= 3 + 5(4) = 23 \\ y &= 10(4) = 40 \\ z &= 4 \end{aligned}$$

✓ Example 1.2.8: A Two Parameter Set of Solutions

Find the solution to the system

$$\begin{aligned}x + 2y - z + w &= 3 \\x + y - z + w &= 1 \\x + 3y - z + w &= 5\end{aligned}$$

Solution

The augmented matrix is

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 3 \\ 1 & 1 & -1 & 1 & 1 \\ 1 & 3 & -1 & 1 & 5 \end{array} \right]$$

We wish to carry this matrix to row-echelon form. Here, we will outline the row operations used.

Take -1 times the first row and add to the second. Then take -1 times the first row and add to the third. This yields

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 3 \\ 0 & -1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 & 2 \end{array} \right]$$

Now add the second row to the third row and divide the second row by -1 .

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 3 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \quad (1.2.13)$$

This matrix is in row-echelon form and we can see that x and y correspond to pivot columns, while z and w do not. Therefore, we will assign parameters to the variables z and w . Assign the parameter s to z and the parameter t to w . Then the first row yields the equation $x + 2y - s + t = 3$, while the second row yields the equation $y = 2$. Since $y = 2$, the first equation becomes $x + 4 - s + t = 3$ showing that the solution is given by

$$\begin{aligned}x &= -1 + s - t \\y &= 2 \\z &= s \\w &= t\end{aligned}$$

It is customary to write this solution in the form

$$\begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} -1 + s - t \\ 2 \\ s \\ t \end{bmatrix} \quad (1.2.14)$$

📌 Note on RREF Algorithm

- Start with the augmented matrix.
- From left to right, locate the first nonzero column. This is the **pivot column**, and its top entry is the **pivot position**. If needed, swap rows so a nonzero entry is in the pivot position.
- Use row operations to make all entries **below** the pivot equal to zero.
- Repeat steps 2–3 with the remaining submatrix, moving downward and to the right, until no rows remain. At this point the matrix is in **row-echelon form**.
- Scale each row so that the leading entry (pivot) in each nonzero row is **1**.
- Finally, working from right to left, use row operations to create zeros **above** each pivot. The result is the **reduced row-echelon form (RREF)**.

Exercises

In Exercises 1 - 6, state whether the given matrix is in reduced row echelon form, row echelon form only or in neither of those forms.

1. $\left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 3 \end{array} \right]$
2. $\left[\begin{array}{ccc|c} 3 & -1 & 1 & 3 \\ 2 & -4 & 3 & 16 \\ 1 & -1 & 1 & 5 \end{array} \right]$
3. $\left[\begin{array}{ccc|c} 1 & 1 & 4 & 3 \\ 0 & 1 & 3 & 6 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{array} \right]$
4. $\left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 4 & 3 \end{array} \right]$
5. $\left[\begin{array}{ccc|c} 1 & 0 & 4 & 3 \\ 0 & 1 & 3 & 6 \\ 0 & 0 & 0 & 0 \end{array} \right]$
6. $\left[\begin{array}{ccc|c} 1 & 1 & 4 & 3 \\ 0 & 1 & 3 & 6 \end{array} \right]$

In Exercises 7 - 12, the following matrices are in reduced row echelon form. Determine the solution of the corresponding system of linear equations or state that the system is inconsistent.

7. $\left[\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & 7 \end{array} \right]$
8. $\left[\begin{array}{ccc|c} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 20 \\ 0 & 0 & 1 & 19 \end{array} \right]$
9. $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 & 4 \\ 0 & 1 & 0 & 6 & -6 \\ 0 & 0 & 1 & 0 & 2 \\ 1 & 0 & 0 & 3 & 0 \end{array} \right]$
10. $\left[\begin{array}{ccc|c} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & -8 & 1 & 7 \end{array} \right]$
11. $\left[\begin{array}{ccc|c} 0 & 1 & 4 & -3 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 9 & -3 & -3 \end{array} \right]$
12. $\left[\begin{array}{ccc|c} 0 & 1 & -4 & 20 \\ 0 & 0 & 0 & 0 \end{array} \right]$

In Exercises 13 - 26, solve the following systems of linear equations using the techniques discussed in this section. Compare and contrast these techniques with those you used to solve the systems in the Exercises in [Section 8.1](#).

13. $\begin{cases} -5x + y = 17 \\ x + y = 5 \end{cases}$
14. $\begin{cases} x + y + z = 3 \\ 2x - y + z = 0 \\ -3x + 5y + 7z = 7 \end{cases}$
15. $\begin{cases} 4x - y + z = 5 \\ 2y + 6z = 30 \\ x + z = 5 \end{cases}$
16. $\begin{cases} x - 2y + 3z = 7 \\ -3x + y + 2z = -5 \\ 2x + 2y + z = 3 \end{cases}$
17. $\begin{cases} 3x - 2y + z = -5 \\ x + 3y - z = 12 \\ x + y + 2z = 0 \end{cases}$

$$\begin{array}{lcl}
 18. & \left\{ \begin{array}{l} 2x - y + z = -1 \\ 4x + 3y + 5z = 1 \\ 5y + 3z = 4 \end{array} \right. \\
 19. & \left\{ \begin{array}{l} x - y + z = -4 \\ -3x + 2y + 4z = -5 \\ x - 5y + 2z = -18 \end{array} \right. \\
 20. & \left\{ \begin{array}{l} 2x - 4y + z = -7 \\ x - 2y + 2z = -2 \\ -x + 4y - 2z = 3 \end{array} \right. \\
 21. & \left\{ \begin{array}{l} 2x - y + z = 1 \\ 2x + 2y - z = 1 \\ 3x + 6y + 4z = 9 \end{array} \right. \\
 22. & \left\{ \begin{array}{l} x - 3y - 4z = 3 \\ 3x + 4y - z = 13 \\ 2x - 19y - 19z = 2 \end{array} \right. \\
 23. & \left\{ \begin{array}{l} x + y + z = 4 \\ 2x - 4y - z = -1 \\ x - y = 2 \end{array} \right. \\
 24. & \left\{ \begin{array}{l} x - y + z = 8 \\ 3x + 3y - 9z = -6 \\ 7x - 2y + 5z = 39 \end{array} \right. \\
 25. & \left\{ \begin{array}{l} 2x - 3y + z = -1 \\ 4x - 4y + 4z = -13 \\ 6x - 5y + 7z = -25 \end{array} \right. \\
 26. & \left\{ \begin{array}{l} x_1 - x_3 = -2 \\ 2x_2 - x_4 = 0 \\ x_1 - 2x_2 + x_3 = 0 \\ -x_3 + x_4 = 1 \end{array} \right.
 \end{array}$$

27. It's time for another meal at our local buffet. This time, 22 diners (5 of whom were children) feasted for \$162.25, before taxes. If the kids buffet is \$4.50, the basic buffet is \$7.50, and the deluxe buffet (with crab legs) is \$9.25, find out how many diners chose the deluxe buffet.
28. Carl wants to make a party mix consisting of almonds (which cost \$7 per pound), cashews (which cost \$5 per pound), and peanuts (which cost \$2 per pound.) If he wants to make a 10 pound mix with a budget of \$35, what are the possible combinations almonds, cashews, and peanuts? (You may find it helpful to review [Example 8.1.3](#) in [Section 8.1](#).)
29. Find the quadratic function passing through the points $(-2, 1)$, $(1, 4)$, $(3, -2)$
30. At 9 PM, the temperature was 60°F ; at midnight, the temperature was 50°F ; and at 6 AM, the temperature was 70°F . Use the technique in [Example 8.2.3](#) to fit a quadratic function to these data with the temperature, T , measured in degrees Fahrenheit, as the dependent variable, and the number of hours after 9 PM, t , measured in hours, as the independent variable. What was the coldest temperature of the night? When did it occur?
31. The price for admission into the Stitz-Zeager Sasquatch Museum and Research Station is \$15 for adults and \$8 for kids 13 years old and younger. When the Zahlenreich family visits the museum their bill is \$38 and when the Nullsatz family visits their bill is \$39. One day both families went together and took an adult babysitter along to watch the kids and the total admission charge was \$92. Later that summer, the adults from both families went without the kids and the bill was \$45. Is that enough information to determine how many adults and children are in each family? If not, state whether the resulting system is inconsistent or consistent dependent. In the latter case, give at least two plausible solutions.
32. Use the technique in [Example 8.2.3](#) to find the line between the points $(-3, 4)$ and $(6, 1)$. How does your answer compare to the slope-intercept form of the line in [Equation 23](#)?
33. With the help of your classmates, find at least two different row echelon forms for the matrix

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 4 & 12 & 8 \end{array} \right]$$

Answers

1. Reduced row echelon form
2. Neither
3. Row echelon form only

4. Reduced row echelon form
5. Reduced row echelon form
6. Row echelon form only
7. $(-2, 7)$
8. $(-3, 20, 19)$
9. $(-3t + 4, -6t - 6, 2, t)$
for all real numbers t
10. Inconsistent
11. $(8s - t + 7, -4s + 3t + 2, s, t)$
for all real numbers s and t
12. $(-9t - 3, 4t + 20, t)$
for all real numbers t
13. $(-2, 7)$
14. $(1, 2, 0)$
15. $(-t + 5, -3t + 15, t)$
for all real numbers t
16. $(2, -1, 1)$
17. $(1, 3, -2)$
18. Inconsistent
19. $(1, 3, -2)$
20. $(-3, \frac{1}{2}, 1)$
21. $(\frac{1}{3}, \frac{2}{3}, 1)$
22. $(\frac{19}{13}t + \frac{51}{13}, -\frac{11}{13}t + \frac{4}{13}, t)$
for all real numbers t
23. Inconsistent
24. $(4, -3, 1)$
25. $(-2t - \frac{35}{4}, -t - \frac{11}{2}, t)$
for all real numbers t
26. $(1, 2, 3, 4)$
27. This time, 7 diners chose the deluxe buffet.
28. If t represents the amount (in pounds) of peanuts, then we need $1.5t - 7.5$ pounds of almonds and $17.5 - 2.5t$ pounds of cashews.
Since we can't have a negative amount of nuts, $5 \leq t \leq 7$.
29. $f(x) = -\frac{4}{5}x^2 + \frac{1}{5}x + \frac{23}{5}$
30. $T(t) = \frac{20}{27}t^2 - \frac{50}{9}t + 60$. Lowest temperature of the evening $\frac{595}{12} \approx 49.58^\circ\text{F}$ at 12:45 AM.
31. Let x_1 and x_2 be the numbers of adults and children, respectively, in the Zahlenreich family and let x_3 and x_4 be the numbers of adults and children, respectively, in the Nullsatz family. The system of equations determined by the given information is

$$\begin{cases} 15x_1 + 8x_2 = 38 \\ 15x_3 + 8x_4 = 39 \\ 15x_1 + 8x_2 + 15x_3 + 8x_4 = 77 \\ 15x_1 + 15x_3 = 45 \end{cases}$$

We subtracted the cost of the babysitter in E3 so the constant is 77, not 92. This system is consistent dependent and its solution is $(\frac{8}{15}t + \frac{2}{5}, -t + 4, -\frac{8}{15}t + \frac{13}{5}, t)$. Our variables represent numbers of adults and children so they must be whole numbers. Running through the values $t = 0, 1, 2, 3, 4$ yields only one solution where all four variables are whole numbers; $t = 3$ gives us $(2, 1, 1, 3)$. Thus there are 2 adults and 1 child in the Zahlenreichs and 1 adult and 3 kids in the Nullsatzs.

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1.3: Rank and Homogenous Systems

Learning Objectives

- Recognize a homogeneous system of linear equations and identify the trivial solution.
- Determine conditions under which a homogeneous system has nontrivial solutions.
- Understand the concept of the rank of a matrix and how it relates to pivot columns.
- Use the rank of a matrix to analyze the number and type of solutions in a homogeneous system.
- Relate the number of equations and variables to the existence of nontrivial solutions.

Up to this point, we have studied general systems of linear equations and seen how pivots determine whether a system has a unique solution, infinitely many solutions, or no solution. In this section, we will focus on a special type of linear system and explore a crucial characteristic of matrices: how their rank, closely related to the positions of pivots, governs the solutions of the system.

Homogeneous Systems

There is a special type of system which requires additional study. This type of system is called a homogeneous system of equations.

Definition: Homogenous System of Linear Equations

A system of equations is called **homogeneous** if each equation in the system is equal to 0. A homogeneous system has the form

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0 \\&\vdots \\a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= 0\end{aligned}$$

where a_{ij} are scalars and x_i are variables.

It is easy to see that this system is consistent, with the origin as a solution. This solution has a special name, which we introduce in the following definition.

Consider the homogeneous system of equations given by

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0 \\&\vdots \\a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= 0\end{aligned}$$

Then, $x_1 = 0, x_2 = 0, \dots, x_n = 0$ is always a solution to this system. We call this the **trivial solution**.

If the system has a solution in which not all of the $x_1, \dots, x_n$ are equal to zero, then we call this solution **nontrivial**. The trivial solution does not tell us much about the system, as it says that $0 = 0$! Therefore, when working with homogeneous systems of equations, we want to know when the system has a nontrivial solution.

Suppose we have a homogeneous system of m equations, using n variables, and suppose that $n > m$. In other words, there are more variables than equations. Then, it turns out that this system always has a nontrivial solution. Not only will the system have a nontrivial solution, but it also will have infinitely many solutions. It is also possible, but not required, to have a nontrivial solution if $n = m$ and $n < m$.

Consider the following example.

✓ Example 1.3.1: Solutions to a Homogeneous System of Equations

Find the nontrivial solutions to the following homogeneous system of equations

$$\begin{aligned} 2x + y - z &= 0 \\ x + 2y - 2z &= 0 \end{aligned}$$

Solution

Notice that this system has $m = 2$ equations and $n = 3$ variables, so $n > m$. Therefore by our previous discussion, we expect this system to have infinitely many solutions.

The process we use to find the solutions for a homogeneous system of equations is the same process we used in the previous section. First, we construct the augmented matrix, given by

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 1 & 2 & -2 & 0 \end{array} \right]$$

Then, we carry this matrix to its reduced row-echelon form, given below.

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right]$$

The corresponding system of equations is

$$\begin{aligned} x &= 0 \\ y - z &= 0 \end{aligned}$$

Since z is not restrained by any equation, we know that this variable will become our parameter. Let $z = t$ where t is any number. Therefore, our solution has the form

$$\begin{aligned} x &= 0 \\ y &= z = t \\ z &= t \end{aligned}$$

Hence this system has infinitely many solutions, with one parameter t .

Suppose we were to write the solution to the previous example in another form. Specifically,

$$\begin{aligned} x &= 0 \\ y &= 0 + t \\ z &= 0 + t \end{aligned}$$

can be written as

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

Notice that we have constructed a column from the constants in the solution (all equal to 0), as well as a column corresponding to the coefficients on t in each equation. While we will discuss this form of solution more in further chapters, for now consider the column of coefficients of the parameter t . In this case, this is the column $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

There is a special name for this column, which is **basic solution**. The basic solutions of a system are columns constructed from the coefficients on parameters in the solution. We often denote basic solutions by X_1, X_2 etc., depending on how many solutions occur.

Therefore, [Example 1.3.1](#) has the basic solution $X_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$.

We explore this further in the following example.

✓ Example 1.3.2: Basic Solutions of a Homogeneous System

Consider the following homogeneous system of equations.

$$\begin{aligned}x + 4y + 3z &= 0 \\ 3x + 12y + 9z &= 0\end{aligned}$$

Find the basic solutions to this system.

Solution

The augmented matrix of this system and the resulting reduced row-echelon form are

$$\left[\begin{array}{ccc|c} 1 & 4 & 3 & 0 \\ 3 & 12 & 9 & 0 \end{array} \right] \rightarrow \cdots \rightarrow \left[\begin{array}{ccc|c} 1 & 4 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

When written in equations, this system is given by

$$x + 4y + 3z = 0$$

Notice that only x corresponds to a pivot column. In this case, we will have two parameters, one for y and one for z . Let $y = s$ and $z = t$ for any numbers s and t . Then, our solution becomes

$$\begin{aligned}x &= -4s - 3t \\ y &= s \\ z &= t\end{aligned}$$

which can be written as

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -4 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

You can see here that we have two columns of coefficients corresponding to parameters, specifically one for s and one for t . Therefore, this system has two basic solutions! These are

$$X_1 = \begin{bmatrix} -4 \\ 1 \\ 0 \end{bmatrix}, X_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

Rank

Another way to gain insight into the solutions of a homogeneous system is to examine the number of pivots in its associated coefficient matrix. This leads us to the concept of **rank**, which we now define as a measure of the number of linearly independent rows (or columns) in a matrix.

Definition 1.3.3: Rank of a Matrix

Let A be a matrix and consider any reduced row-echelon form of A . Then, the number r of pivot columns of A does not depend on the reduced row-echelon form you choose, and is called the **rank** of A . We denote it by $\text{Rank}(A)$.

Similarly, we could count the number of leading entries in the row echelon form of A to determine the rank of A .

✓ Example 1.3.3: Finding the Rank of a Matrix

Consider the matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 5 & 9 \\ 2 & 4 & 6 \end{bmatrix}$$

What is its rank?

Solution

First, we need to find the reduced row-echelon form of A . Through the usual algorithm, we find that this is

$$\begin{bmatrix} \boxed{1} & 0 & -1 \\ 0 & \boxed{1} & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Here we have two leading entries, or two pivot positions, shown above in boxes. The rank of A is $r = 2$.

Notice that we would have achieved the same answer if we had found the row-echelon form of A instead of the reduced row-echelon form.

Suppose we have a homogeneous system of m equations in n variables, and suppose that $n > m$. From our above discussion, we know that this system will have infinitely many solutions. If we consider the rank of the coefficient matrix of this system, we can find out even more about the solution. Note that we are looking at just the coefficient matrix, not the entire augmented matrix.

Theorem 1.3.1: Rank and Solutions to a Homogeneous System

Let A be the $m \times n$ coefficient matrix corresponding to a homogeneous system of equations, and suppose A has rank r . Then, the solution to the corresponding system has $n - r$ parameters.

Notice that if $n = m$ or $n < m$, it is possible to have either a unique solution (which will be the trivial solution) or infinitely many solutions.

✓ Example 1.3.4: Finding the Rank of a Matrix

Consider the matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 5 & 9 \\ 2 & 4 & 6 \end{bmatrix}$$

What is its rank?

Solution

First, we need to find the reduced row-echelon form of A . Through the usual algorithm, we find that this is

$$\begin{bmatrix} \boxed{1} & 0 & -1 \\ 0 & \boxed{1} & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Here we have two leading entries, or two pivot positions, shown above in boxes. The rank of A is $r = 2$.

Notice that we would have achieved the same answer if we had found the row-echelon form of A instead of the reduced row-echelon form.

We are not limited to homogeneous systems of equations here. The rank of a matrix can be used to learn about the solutions of any system of linear equations. In the previous section, we discussed that a system of equations can have no solution, a unique solution, or infinitely many solutions. Suppose the system is consistent, whether it is homogeneous or not. The following theorem tells us how we can use the rank to learn about the type of solution we have.

Theorem 1.3.2: Rank and Solutions to a Consistent System of Equations

Let A be the $m \times (n + 1)$ augmented matrix corresponding to a consistent system of equations in n variables, and suppose A has rank r . Then

1. the system has a unique solution if $r = n$
2. the system has infinitely many solutions if $r < n$

We will not present a formal proof of this, but consider the following discussions.

1. *No Solution* The above theorem assumes that the system is consistent, that is, that it has a solution. It turns out that it is possible for the augmented matrix of a system with no solution to have any rank r as long as $r > 1$. Therefore, we must know that the system is consistent in order to use this theorem!
2. *Unique Solution* Suppose $r = n$. Then, there is a pivot position in every column of the coefficient matrix of A . Hence, there is a unique solution.
3. *Infinitely Many Solutions* Suppose $r < n$. Then there are infinitely many solutions. There are less pivot positions (and hence less leading entries) than columns, meaning that not every column is a pivot column. The columns which are *not* pivot columns correspond to parameters. In fact, in this case we have $n - r$ parameters.

Exercises

? Exercise 1

Find the rank of the following matrix.

$$\begin{bmatrix} 4 & -16 & -1 & -5 \\ 1 & -4 & 0 & -1 \\ 1 & -4 & -1 & -2 \end{bmatrix}$$

Answer

Rank = 2

? Exercise 2

Find the rank of the following matrix.

$$\begin{bmatrix} 3 & 6 & 5 & 12 \\ 1 & 2 & 2 & 5 \\ 1 & 2 & 1 & 2 \end{bmatrix}$$

? Exercise 3

Find the rank of the following matrix.

$$\begin{bmatrix} 0 & 0 & -1 & 0 & 3 \\ 1 & 4 & 1 & 0 & -8 \\ 1 & 4 & 0 & 1 & 2 \\ -1 & -4 & 0 & -1 & -2 \end{bmatrix}$$

Answer

Rank = 3

? Exercise 4

Find the rank of the following matrix.

$$\begin{bmatrix} 4 & -4 & 3 & -9 \\ 1 & -1 & 1 & -2 \\ 1 & -1 & 0 & -3 \end{bmatrix}$$

? Exercise 5

Find the rank of the following matrix.

$$\begin{bmatrix} 2 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 7 \\ 1 & 0 & 0 & 1 & 7 \end{bmatrix}$$

Answer

Rank = 3

? Exercise 6

Find the rank of the following matrix.

$$\begin{bmatrix} 4 & 15 & 29 \\ 1 & 4 & 8 \\ 1 & 3 & 5 \\ 3 & 9 & 15 \end{bmatrix}$$

? Exercise 7

Find the rank of the following matrix.

$$\begin{bmatrix} 0 & 0 & -1 & 0 & 1 \\ 1 & 2 & 3 & -2 & -18 \\ 1 & 2 & 2 & -1 & -11 \\ -1 & -2 & -2 & 1 & 11 \end{bmatrix}$$

Answer

Rank = 3

? Exercise 8

Find the rank of the following matrix.

$$\begin{bmatrix} 1 & -2 & 0 & 3 & 11 \\ 1 & -2 & 0 & 4 & 15 \\ 1 & -2 & 0 & 3 & 11 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

? Exercise 9

Find the rank of the following matrix.

$$\begin{bmatrix} -2 & -3 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \\ -3 & 0 & -3 \end{bmatrix}$$

Answer

Rank = 2

? Exercise 10

Find the rank of the following matrix.

$$\begin{bmatrix} 4 & 4 & 20 & -1 & 17 \\ 1 & 1 & 5 & 0 & 5 \\ 1 & 1 & 5 & -1 & 2 \\ 3 & 3 & 15 & -3 & 6 \end{bmatrix}$$

? Exercise 11

Find the rank of the following matrix.

$$\begin{bmatrix} -1 & 3 & 4 & -3 & 8 \\ 1 & -3 & -4 & 2 & -5 \\ 1 & -3 & -4 & 1 & -2 \\ -2 & 6 & 8 & -2 & 4 \end{bmatrix}$$

? Exercise 12

Suppose A is an $m \times n$ matrix. Explain why the rank of A is always no larger than $\min(m, n)$.

Answer

It is because you cannot have more than $\min(m, n)$ nonzero rows in the reduced row-echelon form. Recall that the number of pivot columns is the same as the number of nonzero rows from the description of this reduced row-echelon form.

? Exercise 13

State whether each of the following sets of data are possible for the matrix equation $AX = B$. If possible, describe the solution set. That is, tell whether there exists a unique solution, no solution or infinitely many solutions. Here, $[A|B]$ denotes the augmented matrix.

- A is a 5×6 matrix, $\text{rank}(A) = 4$ and $\text{rank}[A|B] = 4$.
- A is a 3×4 matrix, $\text{rank}(A) = 3$ and $\text{rank}[A|B] = 2$.
- A is a 4×2 matrix, $\text{rank}(A) = 4$ and $\text{rank}[A|B] = 4$.
- A is a 5×5 matrix, $\text{rank}(A) = 4$ and $\text{rank}[A|B] = 5$.
- A is a 4×2 matrix, $\text{rank}(A) = 2$ and $\text{rank}[A|B] = 2$.

Answer

- This says that both A and $A|B$ have exactly one row of zeros at the bottom. This means a solution exists. Since A has 5 columns but only 4 pivots, the system will have at least one free variable, leading to an infinite solution set.
- This surely can't happen. The rank cannot get smaller after augmenting another column since we can't lose a pivot position in doing this.
- This is impossible, since A with only 2 columns cannot possibly have 4 pivots.
- Since $A|B$ has one more pivot than A , in the reduced row echelon form, $A|B$'s last row must be $[0 \ 0 \ 0 \ 0 \ 0 \ 1]$. In this case, there is no solution to the system of equations represented by the augmented matrix.
- In this case, there is a unique solution since the columns of A are independent. In the reduced row echelon form of both A and $A|B$, the last two rows are rows of 0s.

? Exercise 14

Consider the system $-5x + 2y - z = 0$ and $-5x - 2y - z = 0$. Both equations equal zero and so $-5x + 2y - z = -5x - 2y - z$ which is equivalent to $y = 0$. Does it follow that x and z can equal anything? Notice that when $x = 1$, $z = -4$, and $y = 0$ are plugged in to the equations, the equations do not equal 0. Why?

Answer

These are not legitimate row operations. They do not preserve the solution set of the system.

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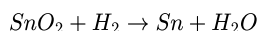
1.4: Chemistry Reaction-Balancing Chemical Reaction

Learning Objectives

- Create a homogenous system of linear equations to balance a chemical equations based on the Mass Balance Theory.
- Understand that the system will have infinitely many solutions and not only the trivial solution.
- Find the solution such that it is the least positive integer that balance the chemical reaction.

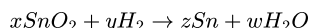
The tools of linear algebra can also be used in the subject area of Chemistry, specifically for balancing chemical reactions.

Consider the chemical reaction



Here the elements involved are tin (Sn), oxygen (O), and hydrogen (H). A chemical reaction occurs and the result is a combination of tin (Sn) and water (H_2O). When considering chemical reactions, we want to investigate how much of each element we began with and how much of each element is involved in the result.

An important theory we will use here is the **Mass Balance Theory**. It tells us that we cannot create or delete elements within a chemical reaction. For example, in the above expression, we must have the same number of oxygen, tin, and hydrogen on both sides of the reaction. Notice that this is not currently the case. For example, there are two oxygen atoms on the left and only one on the right. In order to fix this, we want to find numbers x, y, z, w such that



where both sides of the reaction have the same number of atoms of the various elements.

This is a familiar problem. We can solve it by setting up a system of equations in the variables x, y, z, w . Thus you need

$$\begin{aligned} \text{Sn} : \quad x &= z \\ \text{O} : \quad 2x &= w \\ \text{H} : \quad 2y &= 2w \end{aligned}$$

We can rewrite these equations as

$$\begin{aligned} \text{Sn} : \quad x - z &= 0 \\ \text{O} : \quad 2x - w &= 0 \\ \text{H} : \quad 2y - 2w &= 0 \end{aligned}$$

The augmented matrix for this system of equations is given by

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & 0 & 0 \\ 2 & 0 & 0 & -1 & 0 \\ 0 & 2 & 0 & -2 & 0 \end{array} \right]$$

The reduced row-echelon form of this matrix is

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 \end{array} \right]$$

The solution is given by

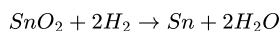
$$\begin{aligned} x - \frac{1}{2}w &= 0 \\ y - w &= 0 \\ z - \frac{1}{2}w &= 0 \end{aligned}$$

which we can write as

$$\begin{aligned} x &= \frac{1}{2}t \\ y &= t \\ z &= \frac{1}{2}t \end{aligned}$$

$$\begin{matrix} \sim & 2 \\ w = t \end{matrix}$$

For example, let $w = 2$ and this would yield $x = 1, y = 2$, and $z = 1$. We can put these values back into the expression for the reaction which yields

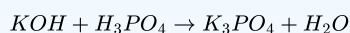


Observe that each side of the expression contains the same number of atoms of each element. This means that it preserves the total number of atoms, as required, and so the chemical reaction is balanced.

Consider another example.

✓ Example 1.4.1: Balancing a Chemical Reaction

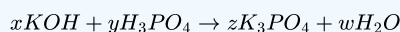
Potassium is denoted by K , oxygen by O , phosphorus by P and hydrogen by H . Consider the reaction given by



Balance this chemical reaction.

Solution

We will use the same procedure as above to solve this problem. We need to find values for x, y, z, w such that



preserves the total number of atoms of each element.

Finding these values can be done by finding the solution to the following system of equations.

$$\begin{aligned} K : & \quad x = 3z \\ O : & \quad x + 4y = 4z + w \\ H : & \quad x + 3y = 2w \\ P : & \quad y = z \end{aligned}$$

The augmented matrix for this system is

$$\left[\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 0 \\ 1 & 4 & -4 & -1 & 0 \\ 1 & 3 & 0 & -2 & 0 \\ 0 & 1 & -1 & 0 & 0 \end{array} \right]$$

and the reduced row-echelon form is

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & -\frac{1}{3} & 0 \\ 0 & 0 & 1 & -\frac{1}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

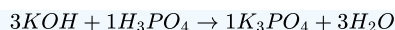
The solution is given by

$$\begin{aligned} x - w &= 0 \\ y - \frac{1}{3}w &= 0 \\ z - \frac{1}{3}w &= 0 \end{aligned}$$

which can be written as

$$\begin{aligned} x &= t \\ y &= \frac{1}{3}t \\ z &= \frac{1}{3}t \\ w &= t \end{aligned}$$

Choose a value for t , say 3. Then $w = 3$ and this yields $x = 3, y = 1, z = 1$. It follows that the balanced reaction is given by



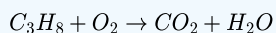
Note that this results in the same number of atoms on both sides.

Of course these numbers you are finding would typically be the number of moles of the molecules on each side. Thus three moles of KOH added to one mole of H_3PO_4 yields one mole of K_3PO_4 and three moles of H_2O .

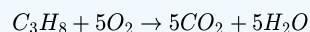
Exercises

? Exercise 1

Balance the following chemical reaction.

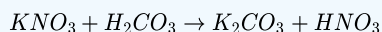


Answer



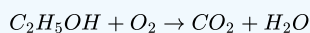
? Exercise 2

Balance the following chemical reaction.

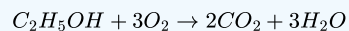


? Exercise 3

Balance the following chemical reaction.

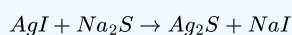


Answer



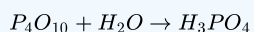
? Exercise 3

Balance the following chemical reaction.

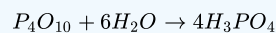


? Exercise 4

Balance the following chemical reaction.



Answer



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1.5: Applications to Physics and Resistor Networks

Learning Objectives

- Understand the parts of an electric circuit.
- Write a system of linear equations to find the value and the direction current in a electric circuits use Kirchhoff's law.
- Solve that system by finding the **RREF** of a augmented matrix to the system.
- Understand that this system should be consistent with a unique solution and interpret the solution.

Resistor Networks

The tools of linear algebra can be used to study the application of resistor networks. An example of an electrical circuit is below.

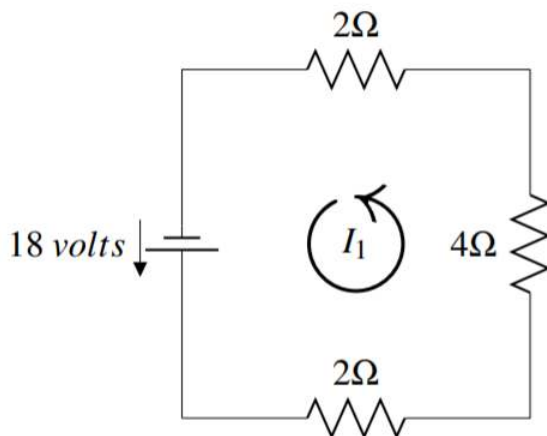


Figure 1.5.1 A circuit with a voltage source of 18 volts that forms a closed rectangular path with three resistors and an arrow indicating current flow. The resistors have resistance 2Ω , 4Ω , and 2Ω , as we move clockwise from the voltage source. The current flow, I_1 , is counterclockwise.

The jagged lines () denote resistors and the numbers next to them give their resistance in ohms, written as Ω . The voltage source () causes the current to flow in the direction from the shorter of the two lines toward the longer (as indicated by the arrow). The current for a circuit is labeled I_k .

In the Figure 1.5.1, the current I_1 has been labeled with an arrow in the counter clockwise direction. This is an entirely arbitrary decision and we could have chosen to label the current in the clockwise direction. With our choice of direction here, we define a positive current to flow in the counter clockwise direction and a negative current to flow in the clockwise direction.

The goal of this section is to use the values of resistors and voltage sources in a circuit to determine the current. An essential theorem for this application is Kirchhoff's law.

Theorem 1.5.1: Kirchhoff's Law

The sum of the resistance (R) times the amps (I) in the counter clockwise direction around a loop equals the sum of the voltage sources (V) in the same direction around the loop.

Kirchhoff's law allows us to set up a system of linear equations and solve for any unknown variables. When setting up this system, it is important to trace the circuit in the counter clockwise direction. If a resistor or voltage source is crossed against this direction, the related term must be given a negative sign.

We will explore this in the next example where we determine the value of the current in the initial diagram.

✓ Example 1.5.1: Solving for Current

Applying Kirchhoff's Law to the circuit in Figure 1.5.1, determine the value for I_1 .

Solution

Begin in the bottom left corner, and trace the circuit in the counter clockwise direction. At the first resistor, multiplying resistance and current gives $2I_1$. Continuing in this way through all three resistors gives $2I_1 + 4I_1 + 2I_1$. This must equal the voltage source in the same direction. Notice that the direction of the voltage source matches the counter clockwise direction specified, so the voltage is positive.

Therefore the equation and solution are given by

$$\begin{aligned} 2I_1 + 4I_1 + 2I_1 &= 18 \\ 8I_1 &= 18 \\ I_1 &= \frac{9}{4} A \end{aligned} \tag{1.5.1}$$

Since the answer is positive, this confirms that the current flows counter clockwise.

✓ Example 1.5.2 Solving for Current

Applying Kirchhoff's Law to the circuit diagram below, determine the value for I_1 .

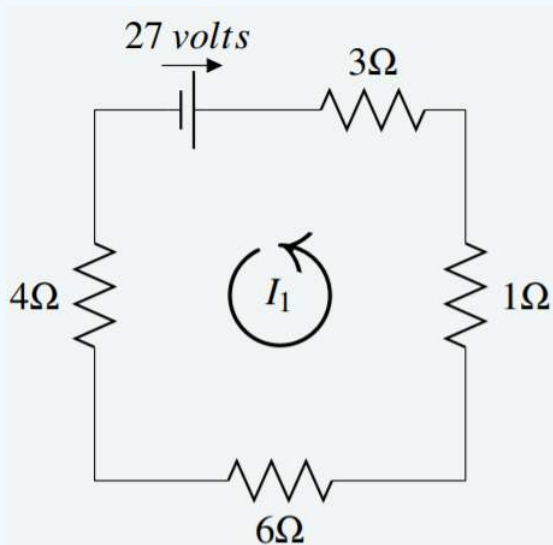


Figure 1.5.2 A circuit with a voltage source of 27 volts that forms a closed rectangular path with four resistors and an arrow indicating current flow. The resistors have resistance 3Ω , 1Ω , 6Ω , and 4Ω , as we move clockwise from the voltage source. The current flow, I_1 , is counterclockwise.

Solution

Begin in the top left corner this time, and trace the circuit in the counter clockwise direction. At the first resistor, multiplying resistance and current gives $4I_1$. Continuing in this way through the four resistors gives $4I_1 + 6I_1 + 1I_1 + 3I_1$. This must equal the voltage source in the same direction. Notice that the direction of the voltage source is opposite to the counter clockwise direction, so the voltage is negative.

Therefore the equation and solution are given by

$$\begin{aligned} 4I_1 + 6I_1 + 1I_1 + 3I_1 &= -27 \\ 14I_1 &= -27 \\ I_1 &= -\frac{27}{14} A \end{aligned} \tag{1.5.2}$$

Since the answer is negative, this tells us that the current flows clockwise.

A more complicated example follows. Two of the circuits below may be familiar; they were examined in the examples above. However as they are now part of a larger system of circuits, the answers will differ.

✓ Example 1.5.3: Unknown Currents

The diagram below consists of four circuits. The current (I_k) in the four circuits is denoted by I_1, I_2, I_3, I_4 . Using Kirchhoff's Law, write an equation for each circuit and solve for each current.

Solution

The circuits are given in the following diagram.

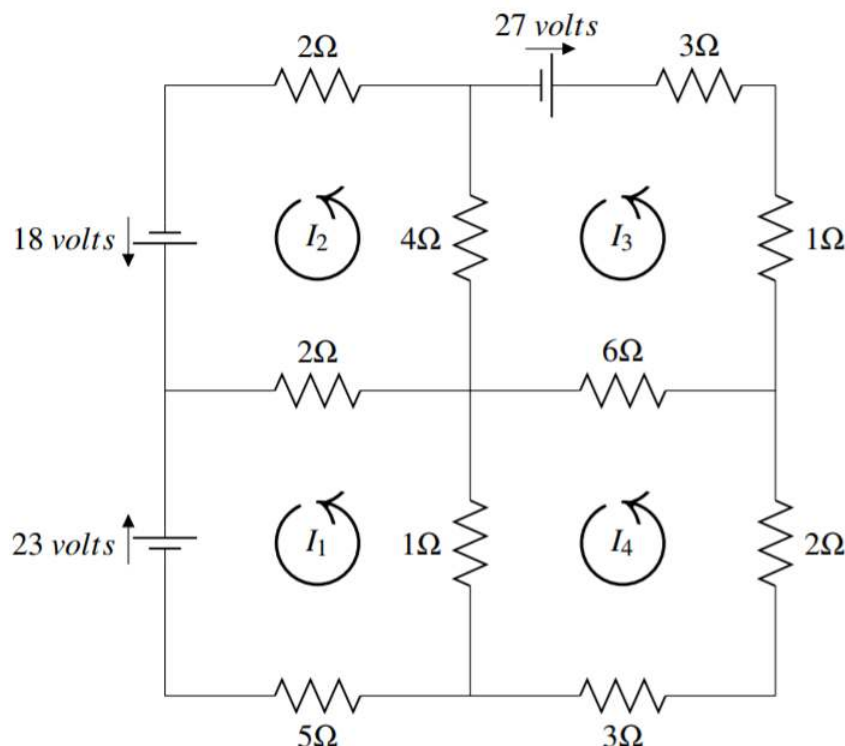


Figure 1.5.3 A circuit with multiple interconnected components, including resistors, voltage sources, and current loops. The circuit is arranged in a 2x2 grid, dividing the circuit into 4 squares: Square 1 on the bottom left, Square 2 on the top left, Square 3 on the top right and Square 4 on the bottom right. Square 1 has counterclockwise current flow I_1 , with a 23V voltage source with arrow pointing up, and resistors 5Ω along the bottom, 1Ω on the right shared with Square 4, and 2Ω above shared with Square 2. Square 2 has counterclockwise current flow I_2 , with an 18V voltage source with arrow pointing down, and resistors 2Ω on the bottom shared with Square 1, 4Ω on the right, shared with Square 3, and 2Ω above. Square 3 has counterclockwise current flow I_3 , with resistors 4Ω on the left shared with Square 2, 6Ω on the bottom shared with Square 4, 1Ω on the right, 3Ω above, and a 27V voltage source with arrow pointing right. Square 4 has counterclockwise current flow I_4 , with resistors 1Ω on the left shared with Square 1, 3Ω below, 2Ω on the right and 6Ω above shared with Square 3.

Starting with the top left circuit, multiply the resistance by the amps and sum the resulting products. Specifically, consider the resistor labeled 2Ω that is part of the circuits of I_1 and I_2 . Notice that current I_2 runs through this in a positive (counterclockwise) direction, and I_1 runs through in the opposite (negative) direction. The product of resistance and amps is then $2(I_2 - I_1) = 2I_2 - 2I_1$. Continue in this way for each resistor, and set the sum of the products equal to the voltage source to write the equation:

$$2I_2 - 2I_1 + 4I_2 - 4I_3 + 2I_2 = 18$$

The above process is used on each of the other three circuits, and the resulting equations are:

Upper right circuit:

$$4I_3 - 4I_2 + 6I_3 - 6I_4 + I_3 + 3I_3 = -27$$

Lower right circuit:

$$2I_1 + 2I_2 + 6I_3 - 6I_4 + I_1 + I_2 = 0$$

$$5I_4 + 4I_4 + 0I_4 - 0I_3 + I_4 - I_1 = 0$$

Lower left circuit:

$$5I_1 + I_1 - I_4 + 2I_1 - 2I_2 = -23$$

Notice that the voltage for the upper right and lower left circuits are negative due to the clockwise direction they indicate.

The resulting system of four equations in four unknowns is

$$\begin{aligned} 2I_2 - 2I_1 + 4I_2 - 4I_3 + 2I_2 &= 18 \\ 4I_3 - 4I_2 + 6I_3 - 6I_4 + I_3 + I_3 &= -27 \\ 2I_4 + 3I_4 + 6I_4 - 6I_3 + I_4 - I_1 &= 0 \\ 5I_1 + I_1 - I_4 + 2I_1 - 2I_2 &= -23 \end{aligned} \quad (1.5.3)$$

Simplifying and rearranging with variables in order, we have:

$$\begin{aligned} -2I_1 + 8I_2 - 4I_3 &= 18 \\ -4I_2 + 14I_3 - 6I_4 &= -27 \\ -I_1 - 6I_3 + 12I_4 &= 0 \\ 8I_1 - 2I_2 - I_4 &= -23 \end{aligned} \quad (1.5.4)$$

The augmented matrix is

$$\left[\begin{array}{cccc|c} -2 & 8 & -4 & 0 & 18 \\ 0 & -4 & 14 & -6 & -27 \\ -1 & 0 & -6 & 12 & 0 \\ 8 & -2 & 0 & -1 & -23 \end{array} \right]$$

The solution to this matrix is

$$\begin{aligned} I_1 &= -3A \\ I_2 &= \frac{1}{4}A \\ I_3 &= -\frac{5}{2}A \\ I_4 &= -\frac{3}{2}A \end{aligned} \quad (1.5.5)$$

This tells us that currents I_1 , I_3 , and I_4 travel clockwise while I_2 travels counter clockwise.

Dimensionless Variables

This section shows how solving systems of equations can be used to determine appropriate dimensionless variables. It is only an introduction to this topic and considers a specific example of a simple airplane wing shown below. We assume for simplicity that it is a flat plane at an angle to the wind which is blowing against it with speed V as shown in Figure 1.5.4.

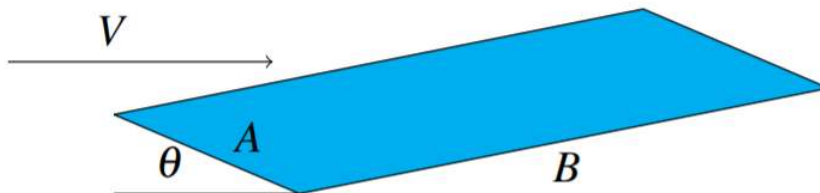


Figure 1.5.4 The airplane, represented as a slanted rectangle with sides A and B , is slanted to the right. The horizontal arrow on the left, labeled V and pointing towards the airplane, indicates wind speed. The slanted side of the rectangle, labeled A , makes an angle θ with the horizontal, whereas the other side of the rectangle, labeled B , is horizontal.

The angle θ is called the angle of incidence, B is the span of the wing and A is called the chord. Denote by l the lift. Then this should depend on various quantities like θ , V , B , A and so forth. Here is a table which indicates various quantities on which it is reasonable to expect l to depend.

Table 1.5.1 Quantities with corresponding symbol and units

Variable	Symbol	Units
chord	A	m
span	B	m
angle incidence	θ	$m^0 kg^0 sec^0$
speed of wind	V	$m sec^{-1}$
speed of sound	V_0	$m sec^{-1}$
density of air	ρ	$kg m^{-3}$
viscosity	μ	$kg sec^{-1} m^{-1}$
lift	l	$kg sec^{-2} m$

Here m denotes meters, sec refers to seconds and kg refers to kilograms. All of these are likely familiar except for μ , which we will discuss in further detail now.

Viscosity is a measure of how much internal friction is experienced when the fluid moves. It is roughly a measure of how “sticky” the fluid is. Consider a piece of area parallel to the direction of motion of the fluid. To say that the viscosity is large is to say that the tangential force applied to this area must be large in order to achieve a given change in speed of the fluid in a direction normal to the tangential force. Thus

$$\mu (\text{area}) (\text{velocity gradient}) = \text{tangential force}$$

Hence

$$(\text{units on } \mu) m^2 \left(\frac{m}{sec\ m} \right) = kg\ sec^{-2} m$$

Thus the units on μ are

$$kg\ sec^{-1} m^{-1}$$

as claimed above.

Returning to our original discussion, you may think that we would want

$$l = f(A, B, \theta, V, V_0, \rho, \mu)$$

This is very cumbersome because it depends on seven variables. Also, it is likely that without much care, a change in the units such as going from meters to feet would result in an incorrect value for l . The way to get around this problem is to look for l as a function of dimensionless variables multiplied by something which has units of force. It is helpful because first of all, you will likely have fewer independent variables and secondly, you could expect the formula to hold independent of the way of specifying length, mass and so forth. One looks for

$$l = f(g_1, \dots, g_k) \rho V^2 AB$$

where the units on $\rho V^2 AB$ are

$$\frac{kg}{m^3} \left(\frac{m}{sec} \right)^2 m^2 = \frac{kg \times m}{sec^2}$$

which are the units of force. Each of these g_i is of the form

$$A^{x_1} B^{x_2} \theta^{x_3} V^{x_4} V_0^{x_5} \rho^{x_6} \mu^{x_7} \quad (1.5.6)$$

and each g_i is independent of the dimensions. That is, this expression must not depend on meters, kilograms, seconds, etc. Thus, placing in the units for each of these quantities, one needs

$$m^{x_1} m^{x_2} (m^{x_4} sec^{-x_4}) (m^{x_5} sec^{-x_5}) (kg m^{-3})^{x_6} (kg sec^{-1} m^{-1})^{x_7} = m^0 kg^0 sec^0$$

Notice that there are no units on θ because it is just the radian measure of an angle. Hence its dimensions consist of length divided by length, thus it is dimensionless. Then this leads to the following equations for the x_i .

$$\begin{aligned} m : \quad & x_1 + x_2 + x_4 + x_5 - 3x_6 - x_7 = 0 \\ \text{sec} : \quad & -x_4 - x_5 - x_7 = 0 \\ kg : \quad & x_6 + x_7 = 0 \end{aligned}$$

The augmented matrix for this system is

$$\left[\begin{array}{ccccccc|c} 1 & 1 & 0 & 1 & 1 & -3 & -1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{array} \right]$$

The reduced row-echelon form is given by

$$\left[\begin{array}{ccccccc|c} 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{array} \right]$$

and so the solutions are of the form

$$\begin{aligned} x_1 &= -x_2 - x_7 \\ x_3 &= x_3 \\ x_4 &= -x_5 - x_7 \\ x_6 &= -x_7 \end{aligned} \tag{1.5.7}$$

Thus, in terms of vectors, the solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \end{bmatrix} = \begin{bmatrix} -x_2 - x_7 \\ x_2 \\ x_3 \\ -x_5 - x_7 \\ x_5 \\ -x_7 \\ x_7 \end{bmatrix}$$

Thus the free variables are x_2, x_3, x_5, x_7 . By assigning values to these, we can obtain dimensionless variables by placing the values obtained for the x_i in the formula (1.5.6). For example, let $x_2 = 1$ and all the rest of the free variables are 0. This yields

$$x_1 = -1, x_2 = 1, x_3 = 0, x_4 = 0, x_5 = 0, x_6 = 0, x_7 = 0$$

The dimensionless variable is then $A^{-1}B^1$. This is the ratio between the span and the chord. It is called the aspect ratio, denoted as AR . Next let $x_3 = 1$ and all others equal zero. This gives for a dimensionless quantity the angle θ . Next let $x_5 = 1$ and all others equal zero. This gives

$$x_1 = 0, x_2 = 0, x_3 = 0, x_4 = -1, x_5 = 1, x_6 = 0, x_7 = 0$$

Then the dimensionless variable is $V^{-1}V_0^1$. However, it is written as V/V_0 . This is called the Mach number $\mathcal{M}$.

Finally, let $x_7 = 1$ and all the other free variables equal 0. Then

$$x_1 = -1, x_2 = 0, x_3 = 0, x_4 = -1, x_5 = 0, x_6 = -1, x_7 = 1$$

then the dimensionless variable which results from this is $A^{-1}V^{-1}\rho^{-1}\mu$. It is customary to write it as $Re = (AV\rho)/\mu$. This one is called the Reynold's number. It is the one which involves viscosity. Thus we would look for

$$l = f(Re, AR, \theta, \mathcal{M}) kg \times m / sec^2$$

This is quite interesting because it is easy to vary Re by simply adjusting the velocity or A but it is hard to vary things like μ or ρ . Note that all the quantities are easy to adjust. Now this could be used, along with wind tunnel experiments to get a formula for the lift which would be reasonable. You could also consider more variables and more complicated situations in the same way.

? Exercise 1

Consider the following diagram of three circuits.

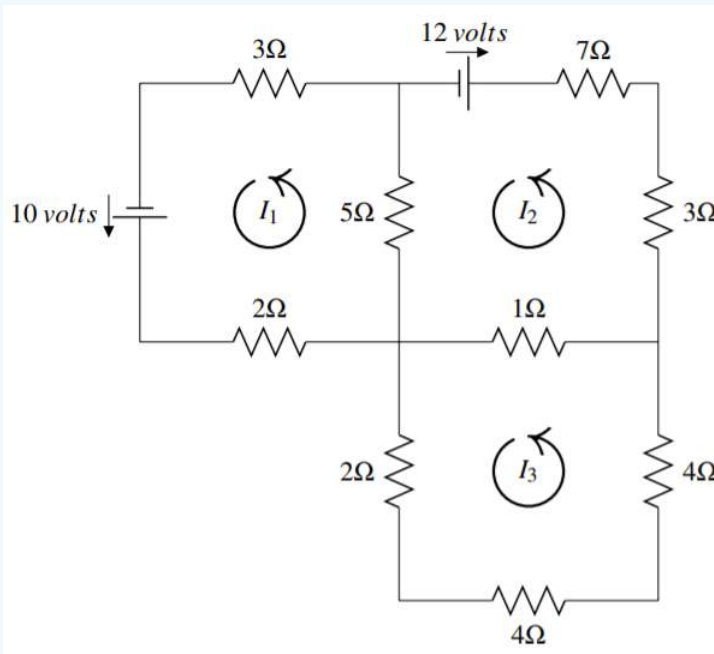


Figure 1.5.2: A circuit with multiple interconnected components, including resistors, voltage sources, and current loops. The circuit is composed of 3 squares: Square 1 on the top left, Square 2 on the top right and Square 3 on the bottom right. Square 1 has counterclockwise current flow I_1 , with a 10V voltage source with arrow pointing down, and resistors 2Ω on the bottom, 5Ω on the right, shared with Square 2, and 3Ω above. Square 2 has counterclockwise current flow I_2 , with resistors 5Ω on the left shared with Square 1, 1Ω on the bottom shared with Square 3, 3Ω on the right, 7Ω above, and a 12V voltage source with arrow pointing right. Square 3 has counterclockwise current flow I_3 , with resistors 2Ω on the left, 4Ω below, 2Ω on the right and 1Ω above shared with Square 2.

The current in amps in the four circuits is denoted by I_1, I_2, I_3 and it is understood that the motion is in the counter clockwise direction. If I_k ends up being negative, then it just means the current flows in the clockwise direction.

Find I_1, I_2, I_3 .

Answer

You have

$$\begin{aligned} 2I_1 + 5I_1 + 3I_1 - 5I_2 &= 10 \\ I_2 - I_3 + 3I_2 + 7I_2 + 5I_2 - 5I_1 &= -12 \\ 2I_3 + 4I_3 + 4I_3 + I_3 - I_2 &= 0 \end{aligned} \tag{1.5.8}$$

Simplifying this yields

$$\begin{aligned} 10I_1 - 5I_2 &= 10 \\ -5I_1 + 16I_2 - I_3 &= -12 \\ -I_2 + 11I_3 &= 0 \end{aligned} \tag{1.5.9}$$

The solution is given by

$$I_1 = \frac{218}{295}, I_2 = -\frac{154}{295}, I_3 = -\frac{14}{295}$$

✓ Exercise 2

Consider a circuit with the following description:

- Left Loop:
 - A 10 V battery
 - A $2\ \Omega$ resistor
 - A shared $3\ \Omega$ resistor
- Right Loop:
 - A 5 V battery
 - A $4\ \Omega$ resistor
 - The same shared $3\ \Omega$ resistor (which lies between the two loops)

Suppose that:

- i_1 is the clockwise loop current in the left loop.
- i_2 is the clockwise loop current in the right loop.
- The current through the shared resistor is given by the difference ($i_1 - i_2$)

Write the equations for each loop and then solve for i_1 and i_2 .

? Exercise 3

Consider a circuit with three loops connected through shared resistors with the following description:

- Loop 1:
 - A 12 V battery
 - A $2\ \Omega$ resistor
 - A shared $4\ \Omega$ resistor with Loop 2
- Loop 2:
 - A 9 V battery
 - A $3\ \Omega$ resistor
 - A shared $4\ \Omega$ resistor with Loop 1
 - A shared $5\ \Omega$ resistor with Loop 3
- Loop 3:
 - A 6 V battery
 - A $2\ \Omega$ resistor
 - A shared $5\ \Omega$ resistor with Loop 2

Currents:

- i_1 flows clockwise in Loop 1
- i_2 flows clockwise in Loop 2
- i_3 flows clockwise in Loop 3

The shared resistors carry the current differences between loops:

- $4\ \Omega$ resistor (Loops 1 & 2): current is $i_1 - i_2$
- $5\ \Omega$ resistor (Loops 2 & 3): current is $i_2 - i_3$

Write the equations for each loop and then solve for i_1 , i_2 and i_3 .

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